

cap-able

Make it more able to caption — now that is cap-able

v0.1.2

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MIT

A comprehensive Typst package for creating professional three-line tables and figures with bilingual caption support, continued tables/figures, subfigures, and flexible customization options for academic documents.

SCHRÖDINGER BLUME

cap-able is a powerful Typst package designed for academic documents, providing:

- **Three-line tables** — Academic-standard tables (top, middle, bottom rules)
- **Bilingual captions** — Auto-formatted bilingual captions (Chinese/English, German/English, etc.)
- **Continued tables/figures** — Multi-page tables and figures with automatic numbering
- **Subfigures** — Flexible multi-image grid layouts with overlay labels and subcaptions
- **Notes** — Auto-width-matched notes for tables and figures
- **Multilingual support** — 25+ languages including RTL languages
- **Unified configuration** — Configure both via `cap-style`, or override per-type with `cap-tab-style` / `cap-fig-style`

The package is particularly suited for academic papers, theses, and technical documents requiring professional typography and multi-language support.

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Part I

Introduction

cap-able brings professional table and figure capabilities to Typst, following academic publishing standards and best practices.

The name “cap-able” is a wordplay on “caption” and “able” — making your documents more able to handle complex captions, very capable, huh!

I.1 Design Philosophy

The package follows these design principles:

1. Declarative Configuration: Configure once at the document start, use everywhere
2. Sensible Defaults: Works out of the box with minimal configuration
3. Language Awareness: Automatic formatting based on document language
4. Academic Standards: Follows established conventions for scholarly publishing

I.2 Dependencies

This package depends on:

- `@preview/tablem:0.3.0` — Markdown-style table syntax support

For documentation generation only:

- `@preview/tidy:0.4.3` — Auto-generate API reference from docstrings
- `@preview/mantys:1.0.2` — Documentation template

Part II

Installation

II.1 From Typst Universe

Once published to Typst Universe, import directly:

```
1 #import "@preview/cap-able:0.1.2": *
```

II.2 Manual Installation

After downloading from the repository:

1. Place the `cap-able` folder in your project directory
2. Import with a relative path:

```
1 #import "cap-able/0.1.2/lib.typ": *
```

Part III

Quick Start

This chapter provides a quick overview of the most common use cases.

III.1 Basic Three-Line Table

The simplest way to create a table uses Markdown-like syntax:

```
1 #captab(  
2   caption: [Sample Data],  
3 ) [  
4   | Name   | Age | Score |  
5   | ----- | --- | ----- |  
6   | Alice  | 25  | 95    |  
7   | Bob    | 30  | 87    |  
8 ]
```

Table 1: Sample Data

Name	Age	Score
Alice	25	95
Bob	30	87

The table automatically gets:

- Content-sized width (default width: `auto`; switch to fixed length / ratio via the `width` parameter or `set-table-width(...)` — see “Table Width” section)
- 1.5pt top rule
- 0.5pt rule below the header
- 1.5pt bottom rule
- Centered content with proper spacing

III.2 Basic Figure

Creating figures is equally straightforward:

```
1 #capfig(  
2   rect(width: 15%, height: 1.15cm, fill: blue.lighten(80%)),  
3   caption: [A Simple Rectangle],  
4 )
```

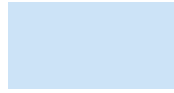


Figure 1: A Simple Rectangle

III.3 Bilingual Captions

For documents requiring bilingual captions, simply add the `caption-en` parameter:

```

1  #set text(lang: "es") // Set the document language to non-English
2
3  #captab(
4    caption: [Parámetros experimentales],
5    caption-en: [Experimental Parameters],
6  )[
7    | Parámetro | Valor | Unidad |
8    | ----- | ---- | ----- |
9    | Temperatura | 25 | °C |
10   | Presión | 1 | atm |
11 ]

```

Tabla 2. Parámetros experimentales
Table 2. Experimental Parameters

Parámetro	Valor	Unidad
Temperatura	25	°C
Presión	1	atm

The package automatically:

- Detects the document language
- Formats the primary caption in that language
- Adds an English caption below when `enable-english-caption: true`

Note: When the document language is English (`#set text(lang: "en")`), if both `caption` and `caption-en` are provided, only `caption-en` is displayed as a single-language caption — `caption` is ignored, since the document is already in English and bilingual output is unnecessary.

Part IV

Tables in Detail

IV.1 Table Syntax

`captab` uses the `tablem` package to parse Markdown-style table syntax, avoiding verbose Typst table syntax.

IV.1.1 Basic Syntax

```
1 | Header1 | Header2 | Header3 |
2 | ----- | ----- | ----- | <- Separator
3 | Cell 1 | Cell 2 | Cell 3 |
4 | Cell 4 | Cell 5 | Cell 6 |
```

Table 3: Basic Syntax Example

Header1	Header2	Header3
Cell 1	Cell 2	Cell 3
Cell 4	Cell 5	Cell 6

Note: The separator row (| --- |) marks the boundary between header and body rows.

IV.1.2 Alignment

Control alignment using colons in the separator:

```
1 | Left | Center | Right |
2 | :----- | :-----: | -----: |
3 | L | C | R |
```

Table 4: Column Alignment Example

Left	Center	Right
L	C	R

IV.1.3 Cell Merging

tablem supports cell merging:

- Use < to merge with cell to the left (horizontal)
- Use ^ to merge with cell above (vertical)

```
1 | A    | B    |
2 | ---- | ---- |
3 | Span | <    |
4 | C    | Span |
5 | D    | ^    |
```

Table 5: Cell Merging Example

A	B
Span	
C	Span
D	

IV.2 Column Configuration

Customize column widths using the columns parameter (the legacy name cols still works as a back-compat alias).

Column width rules:

- Without columns (auto mode): columns fit content; may overflow with long text.
- Only absolute units (e.g., cm, pt): table width = sum of columns; may be narrower or overflow.
- With fr units: table expands proportionally to text width. fr and absolute units can be mixed — absolute columns take fixed widths first, and the remaining space is distributed by fr ratio.

The columns parameter accepts an array of lengths, for example:

```
1 columns: (8cm, 2cm, 2cm)    // Absolute units
2 columns: (3fr, 1fr, 1fr)    // fr units
3 columns: (3fr, 2cm, 1fr)    // Mixed
```

IV.2.1 Auto mode columns: ()

Table 6: Auto Columns

Description	Value	Unit
This is a very very, really really, truly truly long text	42	m/s

IV.2.2 Absolute units columns: (8cm, 3cm, 2cm)

Table 7: Absolute Columns

Description	Value	Unit
This is a very very, really really, truly truly long text	42	m/s

IV.2.3 fr units columns: (3fr, 1fr, 1fr) — **First column third as wide**

Table 8: Proportional Columns

Description	Value	Unit
This is a very very, really really, truly truly long text	42	m/s

IV.2.4 Mixed columns: (5fr, 3cm, 1fr) — **Value column fixed at 3cm; Description and Unit columns share remaining width at 5:1**

Table 9: Mixed Columns

Description	Value	Unit
This is a very very, really really, truly truly long text	42	m/s

IV.3 Additional Lines

Add extra horizontal or vertical lines for complex layouts. Each entry in `hlines` / `vlines` may be an `int` shorthand (just the row/col index, all other fields default) or a full dict:

```
1 // Shorthand: plain int list
2 #captab(hlines: (2, 3), vlines: (1, 2), caption: [...])[ ... ]
3
4 // Full dict (customise stroke / start / end)
5 #captab(
6   hlines: ((row: 3, stroke: 1pt),), // Add 1pt horizontal line after row 3
7   vlines: ((col: 1, start: 1),),    // Add vertical line at column 1 (from row 1)
8   caption: [Table with Extra Lines],
9 )[...]
10
11 // Mixed
12 #captab(hlines: (2, (row: 5, stroke: 1.5pt + red), 7), ...)[ ... ]
```

Table 10: Table with Extra Lines

A	B	C
1	2	3
4	5	6
7	8	9

IV.3.1 Global default stroke extra-rule

When every extra line in a table (or document) shares the same stroke, set a global default rather than repeating it per line:

```
1 #show: captab-style.with(extra-rule: 0.5pt + red)
2
3 #captab(
4   hlines: ((row: 2,), (row: 3,)),    // both inherit 0.5pt + red, no repetition
5 ) [ ... ]
```

`extra-rule` accepts a single value (shared by `hlines` & `vlines`) or a dict form for per-axis control:

```
1 #show: captab-style.with(
2   extra-rule: (h: 1pt + blue, v: 0.3pt + gray),
3 )
```

Per-line stroke still wins:

```
1 #captab(
2   extra-rule: 0.5pt + green,
3   hlines: (
4     (row: 2,),                                // 0.5pt + green ← inherits
5     (row: 5, stroke: 1.5pt + orange), // 1.5pt + orange ← overrides
6   ),
7 )
```

Configurable globally on `cap-style` / `captab-style`, or per-call via `captab(extra-rule: ...)`. Default `0.5pt` matches the previous hard-coded behaviour.

IV.4 Three-Line Strokes

Customize the top, middle and bottom rules of the three-line table via `top-rule` / `middle-rule` / `bottom-rule`. Any Typst stroke value works — thickness, color, dashed:

```
1 #captab(
2   caption: [Custom Rule Styles],
3   top-rule: 2pt + red,           // 2pt red top rule
```

```

4   middle-rule: 0.5pt + gray,      // gray middle rule
5   bottom-rule: stroke(thickness: 2pt, dash: "dashed"), // dashed bottom
6 ) [
7   | A | B | C |
8   | - | - | - |
9   | 1 | 2 | 3 |
10 ]

```

Table 11: Custom Rule Styles

A	B	C
1	2	3

You can also configure them globally via `captab-style`:

```

1 #show: captab-style.with(
2   top-rule: 1.2pt,
3   middle-rule: 0.4pt,
4   bottom-rule: 1.2pt,
5 )

```

`auto` reads from global state (defaults: top/bottom 1.5pt, middle 0.5pt).

IV.5 Disable three-line mode (`three-line-table: false`)

By default `captab` renders a three-line table (manual top/middle/bottom rules, `table stroke: none`). To produce a standard Typst grid table (borders on every cell side), set `three-line-table` to `false`:

```

1 // Per-call
2 #captab(
3   caption: [Standard Typst grid],
4   three-line-table: false,
5 ) [
6   | ID | Name | Value |
7   | -- | ---- | ----- |
8   | 1  | A    | 100    |
9 ]
10
11 // Globally
12 #show: captab-style.with(three-line-table: false)
13 // or
14 #show: cap-style.with(three-line-table: false)

```

When `three-line-table: false`, `top-rule` / `middle-rule` / `bottom-rule` are inactive; user-defined extra `hlines` / `vlines` still apply. Adjust the default Typst grid stroke globally via `set table(stroke: ...)`.

IV.6 Pass-through `tablem` / `table` advanced usage

Apart from `captab`'s own named parameters (`columns` / `cols` / `align` / `size` / `leading` / `inset` / `caption...` / `...-rule` / `breakable` / `repeat...` / `hlines` / `vlines` / `label`, etc.), any other named argument is forwarded as-is to the underlying `table(...)` call, mirroring `tablem`'s advanced-usage surface. Useful pass-throughs:

- `fill`: `color` | `function` (cell background; can be `(x, y)` callback)
- `stroke`: `stroke` | `function` | `dictionary` (per-cell strokes)
- `gutter` / `column-gutter` / `row-gutter`
- `rows`
- any other named parameter that Typst's `table()` accepts

```
1  #let frame(stroke) = (x, y) => (
2    left: if x > 0 { 0pt } else { stroke },
3    right: stroke,
4    top: if y < 2 { stroke } else { 0pt },
5    bottom: stroke,
6  )
7
8  #captab(
9    caption: [Monthly reading list],
10   three-line-table: false,           // full grid instead of three-
    line
11   columns: (0.4fr, 1fr, 1fr),
12   align: left,
13   fill: (_, y) => if calc.odd(y) { rgb("EAF2F5") }, // alternating row fill
14   stroke: frame(rgb("21222C")),      // custom stroke
15 ) [
16   | *Month* | *Title* | *Author* |
17   | ----- | ----- | ----- |
18   | January | The Great Gatsby | F. Scott Fitzgerald |
19   | February | To Kill a Mockingbird | Harper Lee |
20 ]
```

Table 12: Monthly reading list

Month	Title	Author
January	The Great Gatsby	F. Scott Fitzgerald
February	To Kill a Mockingbird	Harper Lee

IV.7 Page Break

Starting with 0.1.0 `captab` allows long tables to break across pages by default. Two relevant parameters:

- `breakable` (bool, default `true`) — whether the table may break. Set to `false` to keep the table atomic on one page (it may overflow).
- `repeat-header` (bool / int, default `true`) — repeat the header row on each continuation page using Typst’s native `table.header(repeat: ...)`. `true` repeats on every continuation page; pass a positive integer `n` to repeat only on the first `n` continuation pages; `false` disables repetition.
- `continued-caption` (bool, default `false`) — whether to render a “Cont. Table X.Y caption” header on each continuation page (same format as `refer-to` mode). The first page still shows the full original caption.

```
1 #captab(
2   caption: [Long Data Table],
3   breakable: true,           // allow page break (default)
4   repeat-header: true,      // repeat header rows (default)
5   continued-caption: true,   // continuation pages show "Cont. Table X.Y"
6 )[
7   | ID | Name | Value |
8   | -- | ---- | ----- |
9   // ...
10 ]
```

Or configure globally:

```
1 #show: captab-style.with(
2   breakable: true,
3   repeat-header: true,
4   continued-caption: true,
5 )
```

Table 13: Long Data Table

ID	Name	Value
001	Apple	12.50
002	Banana	8.30
003	Orange	15.00
004	Grape	22.80
005	Watermelon	35.60
006	Mango	18.90

Continuation of Table 13:
Long Data Table

ID	Name	Value
007	Pineapple	25.40
008	Strawberry	28.70
009	Blueberry	45.20
010	Peach	16.40
011	Pear	10.80
012	Cherry	52.30
013	Pomelo	19.60
014	Dragonfruit	21.50
015	Kiwi	14.30
016	Cantaloupe	32.90
017	Coconut	26.80
018	Lychee	38.40
019	Longan	24.70
020	Mangosteen	48.50

How it works: `continued-caption` does not register a new figure on every continuation page. Each continuation cell uses `query(label)` to locate the main table, reads its number from the counter, and delegates to the `refer-to` branch of `_make_caption_content` to format “Cont. Table X.Y caption ...”. If no `label` was provided, `cap-able` auto-synthesises a hidden `__captab_repeat_<n>` label for internal lookup. The caption row and the header row live in two separate `table.header` blocks (with `level: 1/2`), so `continued-caption: true` can coexist with `repeat-header: false` — the caption repeats while the header does not.

The explicit `refer-to` continuation mechanism (manually-split tables) is still supported and useful when you need to place continuation tables at different positions / chapters.

IV.8 Table Notes

Add explanatory notes below tables:

```

1 #captab(caption: [Statistical Results])[
2   | Variable | Mean | SD   |
3   | ----- | ---- | ---- |
4   | X         | 3.2* | 0.5   |
5   | Y         | 4.1  | 0.8   |
6 ]
7 #captnote[
8   Note:  $p < 0.05$ . SD = Standard Deviation.
9 ]

```

Table 14: Statistical Results

Variable	Mean	SD
X	3.2*	0.5
Y	4.1	0.8

Note: $p < 0.05$. SD = Standard Deviation.

IV.9 Continued Tables

cap-able supports two continuation styles — pick the one that fits the use case:

1. Auto page-break + repeat caption (recommended, since 0.1.0) — let `captab` flow naturally across pages and automatically render “Cont. Table X.Y” on the top of every continuation page.
2. Manual `refer-to` (always available) — split the table into two or more parts and use `refer-to` on the second part to inherit the number. Useful when you want to insert text/figures/other logic between the original and the continuation.

IV.9.1 Auto page-break + repeat caption

Pass `continued-caption: true` to `captab` (`breakable` is `true` by default). `cap-able` uses Typst’s native `table.header(repeat: ...)` to re-emit “Cont. Table X.Y” on the top of every continuation page. The number is anchored to the main table, so no figure is re-registered.

```

1 #captab(
2   caption: [Long Data Table],
3   continued-caption: true,           // continuation pages show "Cont. Table X.Y"
4   // breakable: true,               // already the default
5   // repeat-header: true,           // header also repeats by default
6   label: <tab:long-auto>,
7 )

```



```

8   | ID | Value |
9   | -- | ----- |
10  | 1  | 100   |
11  | 2  | 200   |
12  | 3  | 300   |
13  // ...
14 ]

```

To repeat only the caption but not the markdown header, pass `repeat-header: false`. To let the table flow without any continuation caption (Typst-default behaviour), keep `continued-caption: false` (the default).

When to use: long data tables, fully-automatic page breaks. One line of `continued-caption: true` does it — no manual splitting, no caption duplication.

Known limitation: `continued-caption: true` is incompatible with `caption-position: bottom`. Continuation repetition relies on `table.header(repeat: true)` to inject the caption inside the table; the only mechanism for the bottom edge would be `table.footer`, but that locks the caption inside the table's column boundaries — visually “embedded in the table” — which contradicts what `caption-position: bottom` means (“outside the table, below it”).

When both are set, `cap-able` silently disables `repeat` — equivalent to `continued-caption: false`: the original caption appears once, beneath the last page of the table. To repeat the caption on every page, switch to `caption-position: top`.

IV.9.2 Manual refer-to (preserved)

When you need to insert explanatory text, footnotes or any other content between the original and the continuation, use `refer-to` to split explicitly:

```

1  // First part (original table)
2  #captab(
3    caption: [Long Data Table],
4    label: <tab:long>,           // set label for continuation reference
5  )[
6    | ID | Value |
7    | -- | ----- |
8    | 1  | 100   |
9  ]
10
11 Arbitrary content can go here (notes, smaller table, image, ...).
12
13 // Continuation (header shows "Cont. Table X")

```

```
14 #captab(  
15   caption: [Long Data Table],  
16   refer-to: <tab:long>,      // reference the original for numbering  
17   show-caption: true,        // force the caption text on the continuation  
18 ) [  
19   | ID | Value |  
20   | -- | ----- |  
21   | 2 | 200   |  
22 ]
```

Table 15: Long Data Table

ID	Value
1	100

Arbitrary content can go here (notes, smaller table, image, ...).

Continuation of Table 15: Long Data Table

ID	Value
2	200

Either way, the continuation automatically:

- Uses the same number as the original table
- Adds the continuation prefix/suffix
- Does not create a new entry in the table of contents

IV.9.3 Continuation Mode

`continued-mode` controls the format (applies to both styles):

- "prefix" (default): prefix mode, e.g. "Cont. Table 1"
- "suffix": suffix mode, e.g. "Table 1 (continued)"

Part V

Figures in Detail

V.1 Single Figures

Create single figures with bilingual captions using `capfig`:

```
1 #capfig(  
2   image("example.png", width: 30%),  
3   caption: [Typst Logo],  
4   label: <fig:typst-logo>,  
5 )
```

V.2 Subfigures

`capsubfig` creates multi-image parallel layouts:

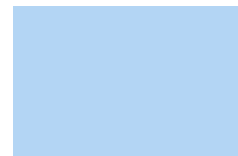
```
1 #capsubfig(  
2   (  
3     (content: rect(width: 3cm, height: 2cm, fill: red.lighten(70%)),  
4       subcaption: [Red]),  
5     (content: rect(width: 3cm, height: 2cm, fill: green.lighten(70%)),  
6       subcaption: [Green]),  
7     (content: rect(width: 3cm, height: 2cm, fill: blue.lighten(70%)),  
8       subcaption: [Blue]),  
9   ),  
10  columns: 3,  
11  show-subcaption: true,      // Show subcaptions  
12  caption: [Color Comparison],  
13 )
```



(a): Red



(b): Green



(c): Blue

Figure 2: Color Comparison

V.3 Overlay Labels

Use overlay label mode to place (a), (b) labels directly on images for a cleaner look:

```
1 #capsubfig(
2   (
3     (content: rect(width: 4cm, height: 3cm, fill: orange.lighten(70%)),
4     (content: rect(width: 4cm, height: 3cm, fill: purple.lighten(70%))),
5   ),
6   columns: 2,
7   caption: [Overlay Labels Demo],
8   label-mode: "overlay",      // Enable overlay mode
9   label-style: "(a)",        // Label style
10  label-bg: white.transparentize(20%), // Semi-transparent white bg
11  label-offset: (5pt, 5pt),   // Offset from top-left
12 )
```



Figure 3: Overlay Labels Demo

V.3.1 Label Style Reference

The `label-style` parameter supports various formats:

Style	Example	Description
"(a)"	(a), (b), (c)	Parenthesized lowercase
"(A)"	(A), (B), (C)	Parenthesized uppercase
"(1)"	(1), (2), (3)	Parenthesized numbers
"(i)"	(i), (ii), (iii)	Parenthesized roman lowercase
"(I)"	(I), (II), (III)	Parenthesized roman uppercase
"a)"	a), b), c)	Trailing parenthesis only
"Fig. A"	Fig. A, Fig. B	English prefix

V.3.2 Decoupling overlay vs subcaption: dict form of `label-style`

By default the overlay label stamped on the figure (`label-mode: "overlay"`) and the subcaption number prefix share a single `label-style` — pass a string and both stay in sync.

To control them independently, pass a dictionary:

```

1  #capsubfig(
2    caption: [Compact overlay, full subcaption],
3    show-subcaption: true,
4    label-mode: "overlay",
5    label-style: (overlay: "a)", subcaption: "(a)",    // overlay "a)", subcaption "(a)
6    caption"
7    (
8      (content: img1, subcaption: [first sub]),
9      (content: img2, subcaption: [second sub]),
10   ),
11 )

```

In the dict, `overlay` controls the overlay style and `subcaption` controls the subcaption prefix; missing keys fall back to the package default `"(a)"`.

Cross-reference letter source: follows whichever side is actually visible — subcaption (when `show-subcaption: true` and `show-subcaption-label: true`) wins, otherwise the overlay style is used (when `label-mode: "overlay"`); when neither is visible the subcaption style is used as a defensive fallback. This keeps the letter in `@fig:xxx` consistent with what the reader actually sees.

V.3.3 Spacing between subcaption number and body

The gap between the subcaption number prefix (e.g. `"(a)"`) and its body text is controlled by `subcaption-number-title-spacing`:

- Default `auto`: inherits the main caption's `number-title-spacing` (the same separator used for “Figure 1.1 caption” in the package defaults), keeping subcaptions visually consistent with the main caption.
- Configure globally via `capfig-style(subcaption-number-title-spacing: ...)`.
- Override per call via `capsubfig(subcaption-number-title-spacing: ...)`.

```

1  #capsubfig(
2    caption: [per-call separator " — "],
3    show-subcaption: true,
4    label-mode: "overlay",
5    subcaption-number-title-spacing: [ — ],
6    (...),
7  )

```

V.4 Subfigure Cross-References

After setting a `label` for each subfigure, reference them with `@`:

```

1  #capsubfig(
2    (

```

```

3      (content: rect(fill: gray), label: <fig:sub-a>),
4      (content: rect(fill: gray), label: <fig:sub-b>),
5  ),
6  label-mode: "overlay",
7  caption: [Referenced Subfigures],
8  label: <fig:main>,
9  )
10
11 See @fig:sub-a and @fig:sub-b for details.

```

Live demonstration:



Figure 4: Referenced Subfigures

See [Figure 4a](#) and [Figure 4b](#) (overall [Figure 4](#)).

V.4.1 subref-style: keep label-style decorations in @ref

By default subref-style: "letter" — @fig:sub-a renders as Fig. 1a (just the letter).

Set "full" to also keep the label-style decorations in cross-references:

```

1 #capsubfig(
2   ...
3   label-style: "(a)",
4   subref-style: "full",      // ← keep the parentheses in @ref
5 )

```

Renders: @fig:sub-a → Fig. 1(a), matching the subcaption prefix.

Works with Chinese-style prefixes (label-style: "图 a" + "full" → 图 1 图 a), brackets ("[A]" → Fig. 1[A]), etc. In "full" mode, label-sep defaults to empty string since the decorations already separate visually; override label-sep explicitly if you want a different separator.

Configurable globally via capfig-style(subref-style: "full") or per-call on capsubfig.

Part VI

Configuration

VI.1 Unified Configuration

`cap-style` configures shared styles (numbering, caption, language, notes, ...) for both tables and figures at once. A subsequent `captab-style` / `capfig-style` call can override per-type.

```
1  #show: cap-style.with(  
2    numbering-format: "1.1",  
3    use-chapter: true,  
4    caption-weight: "regular",  
5    enable-english-caption: true,  
6  )  
7  
8  // Per-type override still works afterwards:  
9  #show: capfig-style.with(  
10   label-mode: "overlay",  
11   label-style: "(a)",  
12 )
```

VI.2 Table Configuration

Use `captab-style` to configure all table-related settings. Typically called via `#show:` at the document start:

```
1  #show: captab-style.with(  
2    // Numbering  
3    numbering-format: "1.1",      // Chapter.Number format  
4    use-chapter: true,           // Include chapter number  
5  
6    // Caption style  
7    caption-size: 10.5pt,  
8    caption-weight: "regular",    // or "bold"  
9  
10   // Spacing  
11   caption-above: 1em,           // Above caption  
12   caption-below: 0.95em,       // Between caption and table  
13  
14   // Language  
15   lang: auto,                  // Auto-detect  
16   enable-english-caption: true, // Enable English sub-caption
```

```

17
18 // Table content
19 body-size: 10.5pt,
20 cell-inset: (x: 3pt, y: 6.5pt),
21
22 // Continuation
23 continued-mode: "prefix", // "prefix" or "suffix"
24 )

```

VI.3 Figure Configuration

Use `capfig-style` for figure and subfigure settings:

```

1 #show: capfig-style.with(
2 // Numbering
3 numbering-format: "1.1",
4
5 // Subfigure defaults
6 label-mode: "overlay", // Default overlay mode
7 label-style: "(a)",
8 gutter: 1em,
9 subcaption-pos: "bottom",
10
11 // Caption
12 enable-english-caption: true,
13 )

```

VI.4 Table Width

`cap-able` does not enforce a fixed table width by default — `width: auto` lets the table size itself by content (a row like `| A | B | C |` only takes three character-widths, not the full text area). The default is decided by `columns`:

- User did not pass `columns`: defaults to `(auto,) * N`, table is content-sized.
- User passed `columns`: respects the user's spec.
 - ▶ `(1fr, 1fr, 1fr)` → `fr` columns expand, but with no fixed-width parent, behaviour depends on context.
 - ▶ `(8cm, 4cm)` → table is 12cm wide, centered.

To fix the table width, three options:

```

1 // (1) Per call: captab(width: ...)
2 #captab(caption: [Narrow Table], width: 60%)[ ... ]
3 #captab(caption: [Absolute Width], width: 8cm)[ ... ]
4 #captab(caption: [Decimal Percent], width: 80.5%)[ ... ]

```



```
5
6 // (2) Globally via captab-style; subsequent captab calls follow
7 #show: captab-style.with(width: 70%)
8
9 // (3) Globally via set-table-width; everything after that follows
10 #set-table-width(width: 50%)           // new form
11 #set-table-width(percentage: 50)      // legacy form (1-100 int, converted to ratio)
```

When `width != auto`, if the user did not pass `columns`, `cap-able` uses `(1fr,) * N` so the table fills the configured width.

`width` accepts three types:

Type	Description / Examples
<code>auto</code>	Unconstrained (default). Table sizes to content.
<code><length></code>	Absolute width. e.g. 8cm, 120pt.
<code><ratio></code>	Percentage of text width, including decimals. e.g. 80%, 80.5%.

Priority: per-call `captab(width:)` > global state (set by `captab-style` / `set-table-width`) > default `auto`.

```
1 #show: captab-style.with(width: 70%)
2
3 #captab(caption: [Following the global 70%])[ ... ]           // 70% wide
4 #captab(caption: [Override to 50%], width: 50%)[ ... ]       // 50% wide
```

Table 16: Example: 50% wide

A	B
C	D

VI.5 Custom Numbering

`numbering-format` accepts any value Typst’s native `numbering()` would accept — most needs are one-liners.

VI.5.1 String formats

As long as it contains a format placeholder (1 arabic, *a*/*A* letter, *i*/*I* roman), other characters render verbatim:

Format	Renders	Use case
<code>"1"</code>	1, 2, 3, ...	Plain arabic (default)
<code>"(A)"</code>	(A), (B), (C), ...	Letter + parentheses
<code>"App. 1"</code>	App. 1, App. 2, ...	Literal prefix + number

Format	Renders	Use case
"1.1"	1.1, 1.2, 2.1, ...	Chapter.number (needs <code>use-chapter: true</code>)
"I.A"	I.A, I.B, II.A, ...	Roman chapter + letter index
"§1-A"	§1-A, §1-B, ...	Symbol + chapter + letter

With `use-chapter: true`, the last placeholder in the format string is treated as the figure/table's own number; the N-1 preceding ones are heading-level prefixes (matching `=`, `==`, `===` ...). For example, "I.1.1.A" inside `=== 2.3.4` for the 5th table renders as II.3.4.E.

VI.5.2 Function form (advanced)

`numbering-format` also accepts a function — equivalent to Typst's native `numbering(..nums)` `=> ...`, 1, 2):

```
1 #show: captab-style.with(
2   use-chapter: false,
3   numbering-format: (..nums) => {
4     let n = nums.pos().last()
5     [Tab§n]           // → "Tab§1", "Tab§2", ...
6   },
7 )
```

With `use-chapter: true`, the function receives all heading levels + the figure/table number (without slicing — your function decides what to consume):

```
1 #show: captab-style.with(
2   use-chapter: true,
3   numbering-format: (..nums) => {
4     let arr = nums.pos()
5     let chap = arr.slice(0, arr.len() - 1) // chapter chain
6     let n = arr.last()                     // figure/table number
7     [§#chap.map(str).join(".")—#n]        // "§2.3—5"
8   },
9 )
```

VI.5.3 Tables vs figures: separate or unified

```
1 // Fully separate (recommended)
2 #show: captab-style.with(numbering-format: "1.1") // tables: 1.1, 1.2
3 #show: capfig-style.with(numbering-format: "I.1") // figures: I.1, I.2
4
5 // Unified via cap-style
```

```
6 #show: cap-style.with(numbering-format: "1.1")           // both tables & figures: 1.1
```

VI.5.4 Escape hatch: bypass cap-able entirely

cap-able installs `set figure(numbering: ...)` inside `captab-style` / `capfig-style`. If even the function form is not enough, write your own `set figure(...)` after the `cap-style` call to fully replace it:

```
1 #show: cap-style.with(...)
2 #set figure.where(kind: table): set figure(numbering: num => [#sym.section.bold #num])
```

Note: this bypasses cap-able’s bilingual supplement / continued-prefix show rules, so confirm you don’t need those.

VI.6 Fine-grained caption text `caption-text`

cap-able’s caption is not a real `figure.caption` element — to support bilingual layout, the caption is rendered manually as `text(...)` calls. As a result, `show figure.caption: set text(font: ...)` does not work. To change caption font / colour / tracking / etc., use the `caption-text` field.

`caption-text` accepts any parameter Typst’s native `text()` takes (font, fill, tracking, spacing, stretch, ...) and supports both flat and nested forms:

VI.6.1 Flat form — applies to the whole caption

```
1 #show: cap-style.with(
2   caption-text: (font: "Times New Roman", fill: blue),
3 )
```

The whole caption (prefix + number + body) is wrapped uniformly.

VI.6.2 Nested form — per-part overrides

If the dict contains any of `whole` / `prefix` / `supplement` / `number` / `body`, it is treated as nested:

Layer	Scope
whole	Entire caption (outermost default)
prefix	Prefix block (supplement + number)
supplement	Just the supplement word (“Table” / “Figure”)
number	Just the digits (“1” / “1.1” / “II.3.4.E” / ...)
body	Just the body (the user-supplied caption text)

Layers cascade: inner overrides outer. Example:

```
1 #show: cap-style.with(
2   caption-text: (
3     whole: (font: "Helvetica"),           // outermost default
```

```

4     number: (fill: red, weight: "bold"), // overlay: red bold on the digits
5     body:   (font: "SimSun"),           // body uses SimSun
6   ),
7 )

```

Result: supplement is Helvetica; number is red bold Helvetica; body is SimSun.

VI.6.3 Common patterns

```

1 // 1. Whole caption Times
2 caption-text: (font: "Times New Roman")
3
4 // 2. Only the digits red
5 caption-text: (number: (fill: red))
6
7 // 3. Prefix bold Times, body regular SimSun
8 caption-text: (
9   prefix: (font: "Times New Roman", weight: "bold"),
10  body:   (font: "SimSun"),
11 )
12
13 // 4. Supplement monospace, number Times, body Helvetica
14 caption-text: (
15   supplement: (font: "Courier"),
16   number:     (font: "Times New Roman"),
17   body:       (font: "Helvetica"),
18 )

```

VI.6.4 Properties

- Default (`:`) does not affect rendering — fully backward compatible.
- When a key collides with `caption-size` / `caption-weight`, the dict's value wins (it acts as an override layer).
- Exposed on `cap-style` / `captab-style` / `capfig-style`.
- Applies to continuation captions too (both `continued-caption: true` and the `refer-to` continuation form).

VI.7 Bilingual Outline

By default, only the primary language caption appears in the outline (table of contents). Enable `outline-bilingual` to show both languages in the outline.

```

1 #show: captab-style.with(
2   outline-bilingual: true,           // Enable bilingual outline
3   outline-separator: " / ",        // Separator between primary and English
4   outline-newline: false,          // false: same line; true: English on new line

```

5)

Three related parameters:

- `outline-bilingual`: Show bilingual captions in outline (default `false`)
- `outline-separator`: Separator between primary and English captions (default `" / "`)
- `outline-newline`: Put English caption on a new line (default `false`)

All three configuration functions (`captab-style`, `capfig-style`, `cap-style`) support these parameters. Use `cap-style` to enable bilingual outline for both tables and figures at once.

Part VII

Multilingual Support

The package supports 25+ languages with automatic localization. Just set the document language and the package automatically uses the correct text.

```
1 #set text(lang: "de") // Switch to German
2
3 #captab(
4   caption: [Experimentelle Ergebnisse],
5   caption-en: [Experimental Results],
6 )[...]
7 // Caption shows: Tabelle 1 Experimentelle Ergebnisse
8 //                Table 1 Experimental Results
```

VII.1 Supported Languages

Code	Language	Code	Language
en	English	zh	Chinese
de	German	fr	French
es	Spanish	it	Italian
pt	Portuguese	ru	Russian
ja	Japanese	ko	Korean
ar	Arabic ↵	he	Hebrew ↵
fa	Persian ↵	ur	Urdu ↵
nl	Dutch	pl	Polish
cs	Czech	sv	Swedish
da	Danish	no	Norwegian
fi	Finnish	tr	Turkish
el	Greek	hi	Hindi
th	Thai	vi	Vietnamese

↵ = Right-to-left language; Traditional Chinese is distinguished via the `region` parameter.

VII.2 Simplified vs Traditional Chinese

The package distinguishes Simplified Chinese (zh) from Traditional Chinese (zh-TW) using Typst's `text.region` parameter:

```
1 // Simplified Chinese (default)
2 #set text(lang: "zh")
3
4 // Traditional Chinese
5 #set text(lang: "zh", region: "TW")
```

Key differences:

Item	Simplified	Traditional
Table prefix	表	表
Figure prefix	图	圖
Continued table	续表	續表
Continued figure	续图	續圖
Continued suffix	(续)	(續)

表 17 繁體中文測試表格
Table 17 Traditional Chinese Test Table

項目	數值
測試	100

續表 17 繁體中文測試表格
Continuation of Table 17 Traditional Chinese Test Table

項目	數值
測試	100

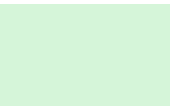


圖 5 繁體中文測試圖片
Figure 5 Traditional Chinese Test Figure

VII.3 RTL Language Support

For RTL languages (Arabic, Hebrew, Persian, Urdu), the package automatically:

- Reverses bilingual caption line order (RTL first, then English)
- Uses `box[]` wrapping to prevent direction confusion in RTL content

- Forces `dir: ltr` for the English line

جدول 18: أهم المدن في العالم العربي
Table 18: Major Cities in the Arab World

المدينة	الدولة	عدد السكان (مليون)
القاهرة	مصر	20.9
بغداد	العراق	7.5
الرياض	المملكة العربية السعودية	7.2
الخرطوم	السودان	5.8
الدار البيضاء	المغرب	3.7
الجزائر	الجزائر	3.5
دمشق	سوريا	2.5
عمّان	الأردن	4.0
بيروت	لبنان	2.4
تونس	تونس	2.3

VII.4 Language-Specific Formatting

Each language has customized separators and spacing:

- Chinese/Japanese: No space between prefix and number, em-space before title
- French: Space before colon (“Tableau 1 : Title”)
- Spanish/Italian: Dot separator (“Table 1. Title”)
- English/German: Colon+space (“Table 1: Title”)

Part VIII

Complete Usage Reference

This chapter is the authoritative reference listing every public function, every parameter, every accepted value, every alias, and every behavioral detail of capable.

VIII.1 Public API Overview

Function	Category	Purpose
<code>cap-style</code>	Config	Shared style for tables & figures
<code>captab-style</code>	Config	Global table style
<code>capfig-style</code>	Config	Global figure style
<code>set-table-width</code>	Config	Global table width %
<code>captab</code>	Table	Three-line table
<code>capfig</code>	Figure	Single bilingual figure
<code>capsubfig</code>	Figure	Subfigure layout
<code>captnote</code>	Note	Table note
<code>capfnote</code>	Note	Figure note
<code>bicap</code>	Caption	Standalone bilingual caption

VIII.2 `set-table-width` Complete Parameters

Param	Type	Description
<code>width</code>	<code>auto</code> / <code>length</code> / <code>ratio</code>	New API. <code>auto</code> = unconstrained; pass <code>8cm</code> / <code>80%</code> / <code>80.5%</code>
<code>percentage</code>	<code>int</code> (1–100)	Legacy API (kept for backward compat). Converted to <code><int>%</code>

When both are passed, `width` wins.

```
1 #set-table-width(width: 80%)           // new form
2 #set-table-width(width: 8cm)           // absolute width
3 #set-table-width(width: auto)           // back to unconstrained
4 #set-table-width(percentage: 80)        // legacy still works
```

VIII.3 `cap-style` Complete Parameters

`cap-style` is the unified entry point: it writes all shared parameters into both the table and figure config states (equivalent to calling `captab-style.with(...)` and `capfig-`

`style.with(...)` in sequence). A subsequent `capstab-style / capfig-style` call can still override per-type.

Param	Default	Description
<code>numbering-format</code>	<code>"1"</code>	Numbering format: any string accepted by Typst's native <code>numbering()</code> (e.g. <code>"1.1"</code> , <code>"A.1"</code> , <code>"§1-A"</code>) or a function (<code>..nums</code>) => <code>content</code>
<code>use-chapter</code>	<code>false</code>	Include chapter number
<code>supplement</code>	<code>auto</code>	Main language prefix
<code>supplement-en</code>	<code>auto</code>	English prefix
<code>continued-prefix</code>	<code>auto</code>	Continued prefix (main lang)
<code>continued-prefix-en</code>	<code>auto</code>	Continued prefix (English)
<code>continued-suffix</code>	<code>auto</code>	Continued suffix (main lang)
<code>continued-suffix-en</code>	<code>auto</code>	Continued suffix (English)
<code>continued-mode</code>	<code>"prefix"</code>	Continuation mode <code>"prefix"</code> / <code>"suffix"</code>
<code>continued-show-caption</code>	<code>auto</code>	Show caption text in continuation
<code>caption-size</code>	<code>10.5pt</code>	Caption size
<code>caption-weight</code>	<code>"regular"</code>	Caption weight
<code>caption-text</code>	<code>(:)</code>	Pass-through dict for caption <code>text(...)</code> . Flat form (e.g. <code>(font: "Times")</code>) applies to the whole caption; nested form (any of <code>whole</code> / <code>prefix</code> / <code>supplement</code> / <code>number</code> / <code>body</code>) layers per part. See “Fine-grained caption text” section.
<code>caption-leading</code>	<code>0.5em</code>	Caption line spacing
<code>caption-above</code>	<code>1.5em</code>	Space above caption
<code>pre-supplement-number-spacing</code>	<code>auto</code>	Prefix–number spacing
<code>post-supplement-number-spacing</code>	<code>auto</code>	Number–suffix spacing
<code>number-title-spacing</code>	<code>auto</code>	Number–title separator
<code>number-title-spacing-en</code>	<code>auto</code>	Number–English–title separator
<code>lang</code>	<code>auto</code>	Language code override
<code>enable-english-caption</code>	<code>true</code>	Enable English sub-caption
<code>outline-bilingual</code>	<code>false</code>	Bilingual outline
<code>outline-separator</code>	<code>" / "</code>	Outline separator
<code>outline-newline</code>	<code>false</code>	Newline in outline
<code>after-indent</code>	<code>auto</code>	Post-block indent fix
<code>note-above</code>	<code>0.7em</code>	Above note
<code>note-below</code>	<code>1em</code>	Below note
<code>note-size</code>	<code>10.5pt</code>	Note size
<code>note-leading</code>	<code>6.5pt</code>	Note leading

Param	Default	Description
<code>note-justify</code>	<code>true</code>	Justify note
<code>caption-position</code>	<code>auto</code>	Caption position in <code>#bicap()[body]</code> mode (applies to both <code>table/figure</code> ; <code>auto</code> keeps each kind's default)
<code>caption-align</code>	<code>auto</code>	Caption horizontal alignment: string <code>"center"</code> / <code>"left"</code> / <code>"right"</code> / <code>"text-left"</code> / <code>"text-right"</code> , or dict (<code>main</code> , <code>continued</code>) to split
<code>placement</code>	<code>none</code>	Floating placement: <code>none</code> (in-flow) / <code>top</code> / <code>bottom</code> (float top/bottom of page) / <code>auto</code> (Typst picks the closer of top/bottom); forces <code>breakable: false</code> when active

Parameters not exposed by `cap-style` — **table-only** (`body-size` / `body-leading` / `cell-inset` / `inset` / `table-below` / `width` / `three-line-table` / `top-rule` / `middle-rule` / `bottom-rule` / `breakable` / `repeat-header` / `continued-caption`) and **figure-only** (`figure-above` / `figure-below` / `subcaption-*` / `gutter` / `subfigure label-*`) — must still be configured via `captab-style` / `capfig-style` directly.

VIII.4 *captab-style* Complete Parameters

All parameters of `captab-style` with defaults. `auto` means auto-select based on document language.

Patch semantics: every parameter actually defaults to `auto`, meaning “leave the current state untouched”. The “Default” column below shows the value used the first time the function is called. Subsequent `captab-style.with(...)` calls only override the fields you supply — any field you omit keeps its previous value. This lets you patch one dimension at a time without re-stating the entire configuration.

Param	Default	Description
<code>caption-above</code>	<code>1.5em</code>	Space above caption block
<code>caption-below</code>	<code>0.5em</code>	Between caption and body
<code>table-below</code>	<code>0.5em</code>	Below whole table
<code>caption-leading</code>	<code>0.5em</code>	Caption line spacing
<code>numbering-format</code>	<code>"1"</code>	Numbering format: any string accepted by Typst's native <code>numbering()</code> or a function <code>(..nums) => content</code> . See “Custom numbering” section for details.
<code>use-chapter</code>	<code>true</code>	Include chapter number

Param	Default	Description
supplement	auto	Main language prefix
supplement-en	auto	English prefix
continued-prefix	auto	Continued prefix
continued-prefix-en	auto	English continued prefix
continued-suffix	auto	Continued suffix
continued-suffix-en	auto	English continued suffix
continued-mode	"prefix"	Continuation mode "prefix" / "suffix"
continued-show-caption	auto	Show caption text in continuation
caption-size	10.5pt	Caption size
caption-weight	"regular"	Caption weight
caption-text	(:)	Pass-through dict for caption text(...). Flat form (e.g. (font: "Times")) applies to the whole caption; nested form (any of whole / prefix / supplement / number / body) layers per part. See “Fine-grained caption text” section.
pre-supplement-number-spacing	auto	Prefix–number spacing
post-supplement-number-spacing	auto	Number–suffix spacing
number-title-spacing	auto	Number–title separator
number-title-spacing-en	auto	Number–English–title separator
lang	auto	Language override
enable-english-caption	true	Enable English sub-caption
body-size	10.5pt	Body font size
body-leading	0.45em	Body line spacing
cell-inset	(x: 5pt, y: 5pt)	Cell padding (dict or scalar); alias of inset, cell-inset wins if both passed
inset	auto	Alias of cell-inset (matches captab’s inset parameter name)
note-above	0.5em	Above note
note-below	1em	Below note
note-size	10.5pt	Note size
note-leading	6.5pt	Note leading
note-justify	true	Justify note
outline-bilingual	false	Bilingual outline
outline-separator	" / "	Outline separator
outline-newline	false	Newline in outline
after-indent	auto	Post-table indent fix

Param	Default	Description
<code>width</code>	<code>auto</code>	Table width. <code>auto</code> =unconstrained (content-sized); <code><length></code> like <code>8cm</code> ; <code><ratio></code> like <code>80%</code> / <code>80.5%</code>
<code>three-line-table</code>	<code>true</code>	Three-line table mode (<code>false</code> skips the manual rules and uses Typst's default <code>table()</code> grid stroke)
<code>top-rule</code>	<code>1.5pt</code>	Three-line top rule stroke (any stroke value, e.g. <code>2pt + red</code> ; only used when <code>three-line-table: true</code>)
<code>middle-rule</code>	<code>0.5pt</code>	Three-line middle rule stroke
<code>bottom-rule</code>	<code>1.5pt</code>	Three-line bottom rule stroke
<code>extra-rule</code>	<code>0.5pt</code>	Default stroke for <code>hlines</code> / <code>vlines</code> entries that omit <code>stroke</code> ; accepts a single value or (<code>h: ...</code> , <code>v: ...</code>) dict for per-axis control
<code>breakable</code>	<code>true</code>	Allow the table to break across pages (outer block's <code>breakable</code>)
<code>repeat-header</code>	<code>true</code>	Repeat the markdown header row on each continuation page (<code>true</code> / <code>false</code> / positive int <code>n</code>)
<code>show-continued-caption</code>	<code>false</code>	Repeat the caption on each continuation page using the refer-to ("Cont. Table X.Y") format. Deprecated alias <code>continued-caption</code> still works, removed in 0.2.0
<code>caption-position</code>	<code>top</code>	Position of the caption relative to body in <code>#bicap()[body]</code> mode (<code>top</code> / <code>bottom</code>)
<code>caption-align</code>	<code>"center"</code>	Caption horizontal alignment: <code>"center"</code> / <code>"left"</code> / <code>"right"</code> (<code>table-local</code>) / <code>"text-left"</code> / <code>"text-right"</code> (<code>text-area-local</code>); or dict (<code>main</code> , <code>continued</code>)
<code>placement</code>	<code>none</code>	Floating placement: <code>none</code> / <code>top</code> / <code>bottom</code> / <code>auto</code> (forces <code>breakable: false</code> when active)

Spacing parameters may be length or content: any `*-spacing` or `pre/post-supplement-number-spacing` accepts a length (e.g. `0.5em`) or content directly (e.g. `[\u{3000}]` em-space).

VIII.5 capfig-style Complete Parameters

`capfig-style` merges figure style + figure spacing + subfigure defaults, updating all three states at once.

Patch semantics: same as `captab-style` — every parameter defaults to `auto`; only the fields you supply are written back. The “Default” column shows the initial state-dictionary value.

Caption/numbering/language section

Param	Default	Description
<code>numbering-format</code>	<code>"1"</code>	Numbering format: any string accepted by Typst's native <code>numbering()</code> (e.g. <code>"1.1"</code> , <code>"A.1"</code> , <code>"§1-A"</code>) or a function <code>(.nums) => content</code>
<code>use-chapter</code>	<code>false</code>	Include chapter number
<code>supplement</code>	<code>auto</code>	Main language prefix
<code>supplement-en</code>	<code>auto</code>	English prefix
<code>continued-prefix</code>	<code>auto</code>	Continued prefix (main lang)
<code>continued-prefix-en</code>	<code>auto</code>	Continued prefix (English)
<code>continued-suffix</code>	<code>auto</code>	Continued suffix (main lang)
<code>continued-suffix-en</code>	<code>auto</code>	Continued suffix (English)
<code>continued-mode</code>	<code>"prefix"</code>	Continuation mode <code>"prefix"</code> / <code>"suffix"</code>
<code>continued-show-caption</code>	<code>auto</code>	Show caption text in continuation
<code>caption-size</code>	<code>10.5pt</code>	Caption size
<code>caption-weight</code>	<code>"regular"</code>	Caption weight
<code>caption-text</code>	<code>(:)</code>	Pass-through dict for caption <code>text(...)</code> . Flat form (e.g. <code>(font: "Times")</code>) applies to the whole caption; nested form (any of <code>whole</code> / <code>prefix</code> / <code>supplement</code> / <code>number</code> / <code>body</code>) layers per part. See “Fine-grained caption text” section.
<code>caption-leading</code>	<code>0.5em</code>	Caption line spacing
<code>pre-supplement-number-spacing</code>	<code>auto</code>	Prefix–number spacing
<code>post-supplement-number-spacing</code>	<code>auto</code>	Number–suffix spacing
<code>number-title-spacing</code>	<code>auto</code>	Number–title separator
<code>number-title-spacing-en</code>	<code>auto</code>	Number–English–title separator
<code>lang</code>	<code>auto</code>	Language code override
<code>enable-english-caption</code>	<code>true</code>	Enable English sub-caption
<code>outline-bilingual</code>	<code>false</code>	Bilingual outline

Param	Default	Description
outline-separator	" / "	Outline separator
outline-newline	false	Newline in outline
after-indent	auto	Post-block indent fix
note-above	0.7em	Above note
note-below	1em + 1.5pt	Below note
note-size	10.5pt	Note size
note-leading	6.5pt	Note leading
note-justify	true	Justify note

Figure spacing section

Param	Default	Description
figure-above	1em	Space above figure
figure-below	1em	Space below figure
caption-above	0.5em	Image–caption gap
caption-leading	0.5em	Caption leading
subcaption-above	0.3em	Subfig–subcaption gap
subcaption-below	0.5em	Below subcaption
subcaption-number-title-spacing	auto	Separator between subcaption number and body (auto inherits main caption's number-title-spacing; accepts content / length)

Subfigure defaults section

Param	Default	Description
gutter	1em	Horizontal gap
subcaption-pos	"bottom"	Subcaption position
show-subcaption	false	Show subcaptions
show-subcaption-label	true	Subcaption includes label
align	"horizon"	Vertical alignment
label-mode	none	Label mode (none / "overlay")
label-style	"(a)" / dict	Label style: pass str to share between overlay and subcaption; pass (overlay: ..., subcaption: ...) dict for per-side control (missing keys fall back to "(a)")
label-font	("Arial",)	Label font list
label-size	12pt	Label size
label-offset	(4pt, 4pt)	Offset
label-text-color	black	Text color

Param	Default	Description
label-stroke	none	Stroke
label-bg	none	Background
label-bg-shape	"rect"	Background shape
label-bg-radius	2pt	Rect radius
label-bg-inset	3pt	Background padding
label-sep	auto	Subref separator (<i>auto</i> : number -> ".", letter -> "; defaults to "" in <i>subref-style</i> : "full" mode)
subref-style	"letter"	Subref letter style: "letter" (just the letter, e.g. Fig. 1a) / "full" (with label-style decorations, e.g. Fig. 1(a))
caption-position	bottom	Caption position relative to body in <code>#bicap()[body]</code> mode (figures default to bottom)
caption-align	"center"	Caption horizontal alignment: "center" / "left" / "right" / "text-left" / "text-right"; or dict (<i>main</i> , <i>continued</i>)
placement	none	Floating placement: <i>none</i> / <i>top</i> / <i>bottom</i> / <i>auto</i>

VIII.6 captab Complete Parameters

Param	Type	Description
columns	auto / int / array	Column config (preferred; <i>auto</i> or length array, supports <i>fr</i> / <i>abs</i> units)
cols	auto / int / array	Back-compat alias for <i>columns</i> (<i>columns</i> wins if both set)
align	auto / alignment / array / function	Cell alignment; merged with <i>mark-down</i> :---: syntax (forwarded to <i>tablem</i> then to <i>table.align</i>)
size	auto / length	Body font size
leading	auto / length	Body line spacing
inset	auto / length / dictionary	Cell padding (alias of <i>cell-inset</i> ; <i>cell-inset</i> wins if both passed)
cell-inset	auto / length / dictionary	Alias of <i>inset</i> (matches the global <i>captab-style.cell-inset</i> name)
caption	none / content	Main language caption
caption-en	none / content	English caption
refer-to	none / label	Label ref for continuation
show-caption	auto / bool	Show caption in continuation

Param	Type	Description
width	auto / length / ratio	Table width (auto reads global, default auto unconstrained; pass 8cm / 80% / 80.5%)
caption-position	auto / top / bottom	Caption position (auto reads global caption-position, default top; bottom places the caption below the table; combined with continued-caption: true the latter is silently disabled)
caption-align	auto / str / dict	Caption horizontal alignment: "center" / "left" / "right" (table-local) / "text-left" / "text-right" (text-area-local); or dict (main, continued) to split; auto reads global
placement	none / top / bottom / auto	Floating placement (none in-flow, top/bottom to page edge, auto lets Typst pick the closer of top/bottom); omit to follow global state (initially none); forces breakable: false when active
three-line-table	auto / bool	Three-line mode (auto reads global three-line-table, default true)
top-rule	auto / stroke	Top rule stroke (auto reads global top-rule, default 1.5pt; only used when three-line-table: true)
middle-rule	auto / stroke	Middle rule stroke (auto reads global middle-rule, default 0.5pt)
bottom-rule	auto / stroke	Bottom rule stroke (auto reads global bottom-rule, default 1.5pt)
extra-rule	auto / stroke / dict	Default stroke for hlines / vlines (when entries omit stroke); single value or (h: ..., v: ...) dict for per-axis (missing keys fall back to 0.5pt); per-line stroke still wins
breakable	auto / bool	Allow page break (auto reads global breakable, default true)
repeat-header	auto / bool / int	Repeat header row on continuation pages (auto reads global repeat-header)
show-continued-caption	auto / bool	Repeat caption on each continuation page (refer-to format). Deprecated alias continued-caption still works (removed in 0.2.0); preferred name wins if both passed
hlines	array of int / dictionary	Extra horizontal rules; entries may be int (shorthand for (row: N)) or full dicts
vlines	array of int / dictionary	Extra vertical rules; entries may be int (shorthand for (col: N)) or full dicts

Param	Type	Description
label	none / label	Cross-reference label
content	content (positional)	Markdown-style body

`hlines / vlines` dict structure:

Key	hlines fault	de- vlines fault	de- Description
row / col	2	1	Row above (h) or col right (v)
start	0	0	Start col/row index
end	none	none	End index (none = to end)
stroke	0.5pt	0.5pt	Rule stroke

```
1 #captab(  
2   hlines: (  
3     (row: 2, stroke: 1pt),  
4     (row: 4, start: 1, end: 3, stroke: 0.3pt + red),  
5   ),  
6   vlines: (  
7     (col: 1, start: 1, stroke: 0.5pt),  
8   ),  
9   caption: [Complex Rules],  
10 ) [...]
```

Table 19: Complex Rules

A	B	C	D
1	2	3	4
5	6	7	8
9	0	1	2

VIII.6.1 Continuation Mode Examples

Prefix mode (default): shows "Continuation of Table 1.1"

Suffix mode: shows "Table 1.1 (continued)"

```
1 #show: captab-style.with(continued-mode: "suffix")  
2 #captab(caption: [Long Table], label: <tab:long2>)[...]  
3 #captab(refer-to: <tab:long2>)[...] // Shows "Table 1.1 (continued)"
```

Force show / hide caption text:

```
1 // Force show  
2 #captab(  

```

```

3   refer-to: <tab:original>,
4   caption: [Original Caption],
5   show-caption: true,
6 )[...]
7
8 // Force hide (even when caption is provided)
9 #captab(
10  refer-to: <tab:original>,
11  show-caption: false,
12 )[...]

```

VIII.7 *bicap* Standalone Caption Function

bicap is the standalone caption generator; usable independently of *captab*/*capfig* for custom layouts or embedding bilingual captions in plain Typst tables/figures.

Starting from 0.1.0 *bicap* supports two calling forms:

- **Caption only:** `#bicap(caption: ..., kind: ...)` — renders the caption alone in its own non-breakable block.
- **Caption + body:** `#bicap(caption: ..., kind: ...)[body]` — wraps *body* and the caption together in a single non-breakable block, conceptually a cap-able-flavoured `figure(body, caption: ...)`. The *body* may also be passed via the named argument `body: [...]`.

The relative position of caption vs. *body* is controlled by `caption-position`:

- The default follows the global state — `top` for tables, `bottom` for figures.
- Override globally via `captab-style` / `capfig-style` with `caption-position: top | bottom`.
- Override per-call by passing `position: top | bottom` to *bicap*.

VIII.7.1 Caption horizontal alignment `caption-align`

`caption-align` controls how the caption sits horizontally relative to the table/figure and the text area. Five values:

Value	Meaning
"center"	(default) Caption centered within the table/figure's own width (current behaviour)
"left"	Aligned to the table/figure's left edge (table-local)
"right"	Aligned to the table/figure's right edge (table-local)
"text-left"	Aligned to the text-area left edge (regardless of table/figure width)
"text-right"	Aligned to the text-area right edge

When `width: 100%` the table/figure fills the text width, so "left" $\equiv$ "text-left" and "right" $\equiv$ "text-right" collapse naturally.

Splitting main vs continuation captions — pass a dict:

```
1 #captab(
2   caption-align: (main: "left", continued: "right"),
3   continued-caption: true,
4   ...
5 ) [ ... ]
```

`main` controls the first-page caption; `continued` controls the “Cont. Table X.Y” caption that appears on each break page when `continued-caption: true`. Missing keys fall back to “center”.

Exposed at:

- **Global:** `cap-style(caption-align: ...)`, `captab-style`, `capfig-style`
- **Per-call:** `captab(caption-align: ...)`, `bicap(caption-align: ...)`

Known limitation: continuation `text-left` / `text-right` silently degrade

The continuation caption emitted by `continued-caption: true` is implemented as a `cap-cell` inside `table.header(level: 1, repeat: true)` — structurally locked to the table’s column width, with no way to extend out to the text-area edges. We considered escape hatches via `place()` and `set page(footer: ...)` but each introduced new bugs (unstable coordinates, clobbering user-defined footers), so we picked an honest degradation over unstable magic.

When `caption-align` (or the `continued` side of a dict) is “`text-left`” / “`text-right`”, the continuation side silently falls back to “`left`” / “`right`” (aligned to the table’s column width). The main caption side is unaffected.

Workaround when needed — manual `refer-to:` split the table into segments and write each continuation segment with a separate `captab(refer-to: <main>, caption-align: “text-left”, ...)`. The continuation caption then lives outside the original table at full text width, and all five alignment values work precisely.

VIII.7.2 Floating placement `placement`

Mirrors Typst’s native `figure(placement: ...)`: lifts the table/figure out of the document flow and floats it to the top or bottom of the current page (or the next one), letting body text fill the remaining space. Equivalent to LaTeX’s `[t]` / `[b]`.

Value	Meaning
<code>none</code> (default)	Render in-flow (no float)
<code>top</code>	Float to the top of the current or next page
<code>bottom</code>	Float to the bottom of the current or next page
<code>auto</code>	Let Typst pick <code>top</code> or <code>bottom</code> automatically — whichever edge is closer to the current document position

```

1 #captab(caption: ..., placement: top)[ ... ]    // float top
2 #captab(caption: ..., placement: bottom)[ ... ] // float bottom
3 #capfig(caption: ..., placement: top)[ ... ]
4 #bicap(placement: top, ...)[ body ]
5
6 // Global: every table floats to the top
7 #show: captab-style.with(placement: top)

```

Exposed at:

- **Global:** `cap-style(placement: ...)`, `captab-style`, `capfig-style`
- **Per-call:** `captab` / `capfig` / `capsubfig` / `bicap`

Known limitations:

1. Floats can't break across pages. Typst's `place(float: true)` requires the floated content to fit on a single page, so:
 - When `placement: top / bottom` is active, `cap-able` forces `breakable: false` on the inner block — even if the user passes `breakable: true`.
 - A long table (taller than one page) will overflow / clip when floated. Keep `placement: none` for such tables and let them flow naturally.
2. Incompatible with `continued-caption: true`. Continuation depends on page breaks, which floats forbid; the combination effectively disables `continued-caption` (no continuation pages exist).
3. Cross-references `@ref: cap-able` wraps the hidden figure (counter-registration anchor) inside the float wrapper too, so `@ref` resolves to the float's landing page — matching what the reader sees. This mirrors Typst's native `figure(placement: ...)` behaviour.

Param	Type	Description
<code>caption</code>	<code>none / content</code>	Main caption
<code>caption-en</code>	<code>none / content</code>	English caption
<code>refer-to</code>	<code>none / label</code>	Continuation ref label
<code>label</code>	<code>none / label</code>	Label for cross-ref
<code>kind</code>	<code>"table" / "figure"</code>	Kind (selects counter and config source)
<code>show-caption</code>	<code>auto / bool</code>	Show caption in continuation
<code>config</code>	<code>auto / dictionary</code>	<code>auto</code> reads the global config matching <code>kind</code> ; otherwise uses the dict
<code>position</code>	<code>auto / top / bottom</code>	Caption position relative to body (only when body is given; <code>auto</code> follows global <code>caption-position</code>)

Param	Type	Description
caption-align	auto / str / dict	Caption horizontal alignment: "center" / "left" / "right" / "text-left" / "text-right", or dict (main, continued); auto reads global
placement	none / top / bottom / auto	Floating placement (auto = Typst picks the closer of top/bottom); forces breakable: false when active
breakable	bool	Whether the outer block may break across pages (default true; lets a breakable body flow naturally)
show-continued-caption	auto / bool	When body breaks across pages, repeat the caption above each native #table() in body (default false; only when kind == "table"). Deprecated alias continued-caption still works, removed in 0.2.0
repeat-header	auto / bool / int	Override the repeat setting of the markdown header in body's #table() (auto leaves user's value alone; true/false/int overrides; only when kind == "table")
body	none / content	Optional body; may also be passed via trailing block #bicap()[...]

```

1  // Form 1: caption only (same as 0.0.x)
2  #align(center)[
3    #rect(width: 6cm, height: 3cm, fill: gray.lighten(70%))
4    #bicap(
5      caption: [Custom Figure],
6      caption-en: [Custom Figure],
7      kind: "figure",
8      label: <fig:custom>,
9    )
10 ]
11
12 // Form 2: bicap wraps the body (kind=table; caption defaults to top)
13 #bicap(
14   caption: [Experimental Data],
15   kind: "table",
16   label: <tab:exp>,
17 )[
18   #table(
19     columns: 3,
20     [A], [B], [C],

```

```

21     [1], [2], [3],
22   )
23 ]
24
25 // kind=figure; caption defaults to bottom — use position: top to flip
26 #bicap(
27   caption: [Schematic],
28   kind: "figure",
29   position: top,
30 )[
31   #image("foo.png", width: 6cm)
32 ]
33
34 // Long table that should repeat the caption on each continuation page (table-kind only)
35 #bicap(
36   caption: [Long Data Table],
37   kind: "table",
38   continued-caption: true,
39   label: <tab:long>,
40 )[
41   #table(
42     columns: 4,
43     table.header([ID], [Name], [Value], [Note]),
44     // ... 50+ rows of data
45   )
46 ]
47
48 // Repeat only the caption on continuation pages — turn off markdown header repetition
49 #bicap(
50   caption: [Long Table],
51   kind: "table",
52   continued-caption: true,
53   repeat-header: false,
54 )[
55   #table(
56     columns: 4,
57     table.header([Col1], [Col2], [Col3], [Col4]),
58     // ...
59   )
60 ]
61
62 // Or just disable markdown header repetition on its own, without continued-caption
63 #bicap(

```

```

64   caption: [Short Table],
65   kind: "table",
66   repeat-header: false,
67 ][ #table(columns: 3, table.header([A], [B], [C]), ...) ]

```

Convention: keep at most one `#table()` per `bicap`. When `continued-caption: true`, `bicap` installs a `show table:` rule that injects the SAME caption header into every native `#table()` in body. If you stuff multiple tables under one `bicap`, every table's continuation page will show the same caption — usually not what you want. Use multiple `bicap(...)` calls or manage each table's caption with `captab`.

Note: `bicap` detects `captab`'s own nested level ≥ 2 header structure and skips the injection in that case (so wrapping `captab(continued-caption: true)` inside `bicap` won't produce stacked captions). Still, avoid nesting `captab` inside a `bicap` with `continued-caption` — the semantics get muddled and numbering can drift.

VIII.8 capfig Complete Parameters

Param	Type	Description
content	content (positional)	Figure body
caption	none / content	Main language caption
caption-en	none / content	English caption
label	none / label	Cross-reference label
refer-to	none / label	Continuation ref label
show-caption	auto / bool	Show caption in continuation
figure-above	auto / length	Space above figure (auto = global)
figure-below	auto / length	Space below figure (auto = global)
caption-above	auto / length	Image-caption gap (auto = global)
caption-leading	auto / length	Caption leading (auto = global)

The figure body and caption are always wrapped in a non-breakable block; the caption is always placed below the image; if `capfig-style.lang` is `auto` while `captab-style.lang` is not, the figure inherits the table's language setting.

VIII.9 capsubfig Complete Parameters

VIII.9.1 Subfigure Dict Structure

`subfigs` is an array of dicts; each supports:

Key	Required	Description
content	Yes	Subfigure body

Key	Required	Description
subcaption	Cond.	Subcaption (used when <code>show-subcaption: true</code>)
label	Opt.	Cross-ref label, e.g. Figure 1.1a
label-style-override	Opt.	Per-item style override (overlay), see below

VIII.9.2 label-style-override dict keys

Per-subfigure overrides for the overlay label. Affects the overlay side only; the subcaption does not participate.

Key	Overrides
style	label-style.overlay (or label-style string form) — per-subfig overlay format string, e.g. "[I]", "{1}"
font	label font list
size	label font size
offset	label (dx, dy) offset
text-color	label text color
stroke	label stroke
bg	label background
bg-shape	background shape
bg-radius	rect radius
bg-inset	background padding

style also affects cross-references: when ref follows overlay (i.e. subcaption is hidden), @ref renders the per-subfig letter in that subfig's own format. When subcaption is visible, ref still follows the global subcaption style; the per-subfig style only changes the figure-stamped label.

VIII.9.3 Function Parameters

All params besides subfigs accept auto (inherit global capfig-style):

- columns (auto / int): Columns per row; auto = single row.
- caption / caption-en / label / refer-to / show-caption: Same as capfig.
- gutter, subcaption-pos, show-subcaption, show-subcaption-label, align, label-mode, label-style, label-font, label-size, label-offset, label-text-color, label-stroke, label-bg, label-bg-shape, label-bg-radius, label-bg-inset, label-sep, subref-style: Override the subfigure defaults from capfig-style.
 - label-style accepts str (overlay/subcaption share the style) or (overlay: ..., subcaption: ...) dict for per-side control. Missing keys fall back to the package default "(a)".
- figure-above, figure-below, caption-above, subcaption-above, subcaption-below: Spacing overrides.
- subcaption-number-title-spacing (auto / content / length): separator between subcaption number and body; auto inherits the main caption's number-title-spacing.

VIII.9.4 Label Style Parse Rules

The parser scans `label-style` left-to-right, locates the first format char (`a/A/1/i/I`); everything before is prefix, everything after is suffix. If no format char exists, the whole string becomes prefix with default format `"a"`.

- Roman numerals support 1–20; fallback to Arabic.
- `label-sep auto` rule: number format -> `"."` (e.g. "Figure 1.1.2"), letter/roman format -> `" "` (e.g. "Figure 1.1a").

VIII.9.5 Overlay with Circle Background

```

1  #capsubfig(
2    (
3      (content: rect(width: 4cm, height: 2.5cm, fill: yellow.lighten(70%))),
4      (content: rect(width: 4cm, height: 2.5cm, fill: red.lighten(70%)),
5        label-style-override: (bg: green, text-color: white)),
6    ),
7    columns: 2,
8    label-mode: "overlay",
9    label-style: "(1)",
10   label-bg: blue.lighten(60%),
11   label-bg-shape: "circle",
12   label-size: 10pt,
13   caption: [Circle bg + per-item override],
14 )

```

VIII.9.6 Multi-row Subfigures

Set `columns: N`; overflow auto-wraps, the last row shrinks to the actual count.

```

1  #capsubfig(
2    (
3      (content: rect(width: 2cm, height: 2cm, fill: red)),
4      (content: rect(width: 2cm, height: 2cm, fill: green)),
5      (content: rect(width: 2cm, height: 2cm, fill: blue)),
6      (content: rect(width: 2cm, height: 2cm, fill: orange)),
7      (content: rect(width: 2cm, height: 2cm, fill: purple)),
8    ),
9    columns: 3,           // 3 per row: 3+2
10   label-mode: "overlay",
11   caption: [Five Subfigures],
12 )

```

VIII.10 `captnote` Complete Parameters

Param	Type	Description
width	auto / ratio / length	Width (three modes below)
justify	auto / bool	Justify (auto = global <code>note-justify</code>)
content	content (positional)	Table note body

Three width modes:

- width: auto (default): Match current `captab` width via `table-width-config`.
- width: 80% etc. ratio: Narrow to percentage, centered.
- width: 10cm etc. length: Fixed width, centered.

```

1 #captnote[Auto match table width]
2 #captnote(width: 70%)[70% width centered]
3 #captnote(width: 8cm)[Fixed 8cm]
4 #captnote(justify: false)[No justification]
```

VIII.11 `capfnote` Complete Parameters

Param	Type	Description
width	ratio / length	Default 100%; auto not supported
justify	auto / bool	Justify (auto = global <code>capfig-style.note-justify</code>)
content	content	Figure note body

```

1 #capfig(image("chart.png"), caption: [Results])
2 #capfnote(width: 90%)[
3   Source: 2024 Survey.
4 ]
```

VIII.12 Full Multilingual Rules

The following table lists prefix words and spacing rules for each language (`pre` = pre-supplement-number-spacing, `sep` = number-title-spacing):

lang	table	figure	pre	sep	continued-suffix
en	Table	Figure	" "	": "	(continued)
zh	表	图	0em	U+3000	(续)
zh-TW	表	圖	0em	U+3000	(續)
de	Tabelle	Abbildung	" "	": "	(Fortsetzung)
fr	Tableau	Figure	" "	" : "	(suite)
es	Tabla	Figura	" "	" . "	(continuación)
it	Tabella	Figura	" "	" . "	(continua)

lang	table	figure	pre	sep	continued-suffix
pt	Tabela	Figura	" "	": "	(continuação)
ru	Таблица	Рисунок	" "	". "	(продолжение)
ja	表	図	0em	U+3000	(続き)
ko	표	그림	" "	". "	(계속)
ar	جدول	شكل	" "	": "	(تابع)
nl	Tabel	Figuur	" "	": "	(vervolg)
pl	Tabela	Rysunek	" "	". "	(cd.)
cs	Tabulka	Obrázek	" "	": "	(pokračování)
sv	Tabell	Figur	" "	". "	(forts.)
da	Tabel	Figur	" "	": "	(fortsat)
no	Tabell	Figur	" "	": "	(forts.)
fi	Taulukko	Kuva	" "	". "	(jatkoa)
tr	Tablo	Şekil	" "	". "	(devam)
el	Πίνακας	Σχήμα	" "	". "	(συνέχεια)
he	טבלה	תמונה	" "	": "	(המשך)
hi	तालिका	चित्र	" "	": "	(जारी)
th	ตาราง	รูป	" "	" "	(ต่อ)
vi	Bảng	Hình	" "	". "	(tiếp theo)
fa	جدول	شكل	" "	": "	(ادامه)
ur	جدول	شكل	" "	": "	(جاری)

↵ = Right-to-left language

Auto-triggers post-table indent fix (after-indent: auto) for: zh, ja, ko, fr, vi, th. Any language can override via explicit after-indent: true/false.

RTL bilingual layout: for ar/he/fa/ur, the main-language and English lines are auto-swapped, the English line is wrapped in `#text(dir: ltr)[...]` to prevent direction confusion, and in monolingual mode the main-language body is wrapped in `box[.]`.

Part IX

API Reference

This chapter provides auto-generated documentation from source code docstrings via tidy.

IX.1 Configuration Module (config.typ)

The configuration module provides global state management, multilingual text, utility functions, and the two main configuration functions.

- `cap-style()`
- `capfig-style()`
- `captab-style()`
- `resolve-lang-indent()`
- `set-figure-width()`
- `set-table-width()`

IX.1.1 cap-style

同时为表格和图片设置全局样式。共享参数会同时作用于两者；之后再调用 `captab-style` 或 `capfig-style` 可对某一类进行覆盖。Configure both table and figure global styles at once. Shared params are applied to both; a subsequent `captab-style` / `capfig-style` call can override for one type only.

Parameters

```

cap-style(
  numbering-format,
  use-chapter,
  supplement,
  supplement-en,
  continued-prefix,
  continued-prefix-en,
  continued-suffix,
  continued-suffix-en,
  continued-mode,
  continued-show-caption,
  caption-size,
  caption-weight,
  caption-text,
  caption-leading,
  caption-above,
  pre-supplement-number-spacing,
  post-supplement-number-spacing,
  number-title-spacing,
  number-title-spacing-en,
  lang,
  enable-english-caption,
  outline-bilingual,
  outline-separator,
  outline-newline,
  after-indent,
  note-above,
  note-below,
  note-size,
  note-leading,
  note-justify,
  ,
  caption-position,
  caption-align,
  placement,
  it
) -> content

```

IX.1.2 capfig-style

Configure global figure and subfigure style within the body. 所有参数默认 auto = “保持当前 state 不变”，多次调用做 patch（增量覆盖）。All params default to auto = “keep current state”. Multiple calls patch incrementally.

Parameters

```
capfig-style(
  numbering-format,
  use-chapter,
  supplement,
  supplement-en,
  continued-prefix,
  continued-prefix-en,
  continued-suffix,
  continued-suffix-en,
  continued-mode,
  continued-show-caption,
  caption-size,
  caption-weight,
  caption-text,
  pre-supplement-number-spacing,
  post-supplement-number-spacing,
  number-title-spacing,
  number-title-spacing-en,
  lang,
  enable-english-caption,
  outline-bilingual,
  outline-separator,
  outline-newline,
  after-indent,
  figure-above,
  figure-below,
  caption-above,
  caption-leading,
  subcaption-above,
  subcaption-below,
  subcaption-number-title-spacing,
  gutter,
  subcaption-pos,
  show-subcaption,
  show-subcaption-label,
  align,
  label-mode,
  label-style,
  label-font,
  label-size,
  label-offset,
  label-text-color,
  label-stroke,
  label-bg,
  label-bg-shape,
  label-bg-radius,
  label-bg-inset,
  label-sep,
  subref-style,
  note-above,
  note-below,
  note-size,
  note-leading,
  note-justify,
  caption-position,
  caption-align,
  placement,
  it
```

IX.1.3 **captab-style**

Configure global table style for all cap-table, bicap and table-note calls within the body. 所有参数默认为 auto = “保持当前 state 不变”，因此可以多次调用 captab-style 仅覆盖部分字段 (patch 语义)，而不会把未提供的字段重置为默认值。All params default to auto = “keep current state”. Multiple captab-style calls patch only the supplied fields; omitted fields are preserved.

Parameters

```

captab-style(
  caption-above,
  caption-below,
  table-below,
  caption-leading,
  numbering-format,
  use-chapter,
  supplement,
  supplement-en,
  continued-prefix,
  continued-prefix-en,
  continued-suffix,
  continued-suffix-en,
  continued-mode,
  continued-show-caption,
  caption-size,
  caption-weight,
  caption-text,
  pre-supplement-number-spacing,
  post-supplement-number-spacing,
  number-title-spacing,
  number-title-spacing-en,
  lang,
  enable-english-caption,
  body-size,
  body-leading,
  cell-inset,
  inset,
  note-above,
  note-below,
  note-size,
  note-leading,
  note-justify,
  outline-bilingual,
  outline-separator,
  outline-newline,
  after-indent,
  width,
  three-line-table,
  top-rule,
  middle-rule,
  bottom-rule,
  extra-rule,
  breakable,
  repeat-header,
  show-continued-caption,
  continued-caption,
  ,
  caption-position,
  caption-align,
  placement,
  it
) -> content

```

IX.1.4 resolve-lang-indent

共享语言与缩进修复解析函数。给定含 lang/after-indent 字段的配置， 返回 (lang, should-indent)。供 captab/capfig/captnote/capfnote 复用。Shared language/indent resolution. Given a config dict with lang/after-indent fields, returns (lang, should-indent). Used by captab/capfig/captnote/capfnote.

Parameters

`resolve-lang-indent`(config)

IX.1.5 set-figure-width

Set figure width percentage (1-100) for capfnote auto-width matching.

Parameters

`set-figure-width`(percentage) -> `content`

IX.1.6 set-table-width

Set table width globally. Accepts `auto` (no fixed width), a length (e.g. 8cm) or a ratio (e.g. 80%, 80.5%).

兼容旧 API: `set-table-width(percentge: 80)` 仍然可用, 会被转成 `80%`。 Backward-compat: legacy `set-table-width(percentge: <int>)` is still accepted and converted to `<int>%` internally.

Parameters

```
set-table-width(  
  width,  
  ,  
  percentage  
) -> content
```

IX.2 Caption Module (bicap.typ)

The caption module provides the core bilingual caption generation engine, supporting both main and continuation (continued table/figure) modes.

- `bicap()`

IX.2.1 bicap

独立双语标题函数, 可用于表格 (kind: "table") 或图片 (kind: "figure")。

调用方式 / Usage:

- `#bicap(caption: [...], kind: "...")` —— 仅渲染题注, 按 `caption-above` / `caption-below` 间距独立成块。
- `#bicap(caption: [...], kind: "...")[body]` —— 把 `body` 与题注一起包进一个不换页 `block`, 相当于一个 `figure(kind: ...)` 的 `cap-able` 版本。题注与 `body` 的相对位置由 `caption-position` 决定 (state 默认: `table=top`, `figure=bottom`; 可通过 `captab-style` / `capfig-style` 全局覆盖, 或直接传 `config`)。

Standalone bilingual caption for tables (`kind: "table"`) or figures (`kind: "figure"`).

Calling forms:

- `#bicap(caption: [...], kind: "...")` — caption only, in its own non-breakable block.
- `#bicap(caption: [...], kind: "...")[body]` — wraps body and caption together, conceptually a `cap-able-flavoured figure(kind: ...)`. Caption position is taken from `caption-position` (default top for tables, bottom for figures; configurable via `captab-style` / `capfig-style` or by passing `config`).

Parameters

```
bicap(
  caption: none content,
  caption-en: none content,
  refer-to: none label,
  label: none label,
  kind: str,
  show-caption: auto bool,
  config: auto dictionary,
  position: auto alignment,
  caption-align: auto str dictionary,
  placement: auto none alignment,
  breakable: bool,
  show-continued-caption: auto bool,
  continued-caption,
  repeat-header: auto bool int,
  body: none content,
  ..body-args
) -> content
```

caption none or content

主语言标题文本 Main language caption text

Default: none

caption-en none or content

英文标题文本（双语模式） English caption text (bilingual mode)

Default: none

refer-to `none` or `label`

引用原始表/图的标签（提供此参数则进入续表模式） Reference to the original table/figure label (enables continuation mode)

Default: `none`

label `none` or `label`

此标题的标签（用于交叉引用） Label for this caption (for cross-references)

Default: `none`

kind `str`

内容类型：“table” 或 “figure” Content type: “table” or “figure”

Default: `"table"`

show-caption `auto` or `bool`

续表/图是否显示标题文本（auto=自动根据 `caption` 是否提供决定） Whether to show caption text in continuation (auto = based on whether caption is provided)

Default: `auto`

config `auto` or `dictionary`

样式配置，`auto` 则从全局 `table-style-config` 获取 Style config, `auto` = use global `table-style-config`

Default: `auto`

position `auto` or `alignment`

题注位置（auto = 跟随全局 `caption-position`; `top` 或 `bottom` 直接覆盖） Caption position (auto = follow global `caption-position`; `top` / `bottom` to override)

Default: `auto`

caption-align `auto` or `str` or dictionary

题注水平对齐：字符串 “center” / “left” / “right” / “text-left” / “text-right” 也接受 dict (main: ..., continued: ...) 拆分主与续。auto = 跟随全局 caption-align。Caption horizontal alignment: string “center” / “left” / “right” / “text-left” / “text-right”. Also accepts dict (main, continued) for split. auto = follow global.

Default: `auto`

placement `auto` or `none` or alignment

浮动定位：none（默认，原地）/ top / bottom（浮到当前页或下一页的顶/底）。auto = 跟随全局 placement。启用时强制 breakable: false（Typst float 不允许跨页）。Floating placement; auto follows global. Forces breakable: false when active.

Default: `()`

breakable `bool`

外层 block 是否允许跨页。默认 true，让 body 里本身可跨页的内容（如长 captab、长段落）顺延到下一页；设为 false 则强制把题注 + body 锁在同一页。Whether the outer block may break across pages. Defaults to true so a body that is itself breakable (e.g. a long captab or paragraph) flows naturally; set false to keep caption + body atomic on one page.

Default: `true`

show-continued-caption auto or bool

跨页时是否在 body 里 每张原生

的顶部重复题注。 仅在 `kind == "table"` 时生效；通过 `show table:` 注入一个 **level-1** `table.header(repeat: true)` 实现, 复用与 `captab(continued-caption: true)` 同款的 `query(label) + here().page()` 检测, 续页才显示。

约定: 一个 **bicap** 里只放一张 `#table()`。如果 **body** 里同时有多张表, 所有表都会被注入相同的题注 (因为 **bicap** 把它们视作“同一题注下的多个表”), 通常不是你想要的; 此时请改写成多个 **bicap** 调用, 或用 `captab` 直接管理每张表的题注。

`kind == "figure"` 下此参数为 **no-op** (图片本身不跨页; 如有跨页图请用 `refer-to`)。

Whether to repeat the caption above each native `#table()` inside body when it breaks across pages. Only active when `kind == "table"`. Implemented by injecting a **level-1** `table.header(repeat: true)` via `show table:`, reusing the `query(label) + here().page()` mechanism from `captab(continued-caption: true)` so the repeated caption only renders on continuation pages.

Convention: keep at most one `#table()` per **bicap**. If body contains multiple tables, the same caption is injected into ALL of them — usually not what you want; in that case use multiple **bicap** calls or manage each table's caption with `captab`.

No-op when `kind == "figure"`; for cross-page figures use `refer-to`. -> **bool**

首选名 `show-continued-caption` (与 `show-subcaption` 等命名统一)。 `continued-caption` 是旧名, **0.1.x** 仍兼容, 将在 **0.2.0** 移除。 Preferred name is `show-continued-caption`; `continued-caption` is a deprecated alias kept through **0.1.x** and removed in **0.2.0**.

Default: auto

repeat-header `auto` or `bool` or `int`

跨页时是否重复 `body` 内 `#table()` 的 `markdown` 表头行。 `auto` (默认) = 不干涉用户写在 `table.header(...)` 里的 `repeat` 设定; `true` | `false` | `<int>` = 强制覆盖, 跟 `Typst` 原生 `table.header(repeat: ...)` 一致。仅 `kind == "table"` 生效, 与 `continued-caption` 互相独立。

Whether to repeat the markdown header row of `#table()` inside `body` across pages. `auto` (default) leaves the user's existing `table.header(repeat: ...)` untouched; `true` | `false` | `<int>` overrides it (matching `Typst`'s native semantics). Only active when `kind == "table"`; independent of `continued-caption`.

Default: `auto`

body `none` or `content`

可选的 `body`: 可以用名参 `body: [...]` 或在调用尾部用内容块 `#bicap()[...]`, 后者会通过 `..body-args` 收集进来。两种形式等价。Optional `body` — pass via the named arg `body: [...]` or as a trailing content block `#bicap()[...]` (collected through `..body-args`); the two forms are equivalent.

Default: `none`

IX.3 Table Module (*table.typ*)

The table module provides academic-standard three-line table creation and its aliases.

- `captab()`

IX.3.1 `captab`

创建三线表, 支持 `Markdown` 语法、双语标题、续表和自定义线条。Create a three-line table with `Markdown` syntax, bilingual captions, continuation, and custom lines.

- `columns` (`auto` | `int` | `array`): 列配置 / Column config (推荐用法 / preferred)
- `cols` (`auto` | `int` | `array`): `columns` 的向后兼容别名 / Back-compatible alias for `columns`
- `align` (`auto` | `alignment` | `array` | `function`): 单元格对齐 (与 `tablem` 的 `:—`: 语法合并) / Cell alignment
- `size` (`auto` | `length`): 表格内容字号 / Table content font size
- `leading` (`auto` | `length`): 表格内容行距 / Table content line spacing
- `inset` (`auto` | `length` | `dictionary`): 单元格内边距 / Cell padding
- `caption` (`none` | `content`): 主语言标题 / Main language caption
- `caption-en` (`none` | `content`): 英文标题 / English caption
- `refer-to` (`none` | `label`): 续表引用的原表标签 / Original label for continuation
- `show-caption` (`auto` | `bool`): 续表是否显示标题文本 / Show caption in continuation

- **three-line-table** (auto | bool): 是否使用三线表样式（默认 true；false 时跳过手动三线，使用 Typst 默认 table 边框） / Three-line table mode
- **top-rule** (auto | stroke): 顶线粗细，接受任何 stroke 值（如 1.5pt + red） / Top rule stroke
- **middle-rule** (auto | stroke): 中线粗细 / Middle rule stroke
- **bottom-rule** (auto | stroke): 底线粗细 / Bottom rule stroke
- **breakable** (auto | bool): 表格能否跨页（默认从全局读取，初始 true） / Whether table may break across pages
- **repeat-header** (auto | bool | int): 跨页时重复表头行 / Repeat the markdown header row on page-break
- **continued-caption** (auto | bool): 跨页时是否在每页表头之上重复题注（题注会进入 table.header 内） / Repeat caption on every page
- **hlines** (array): 额外横线，每项为 (row: int, start: int, end: none|int, stroke: stroke) / Extra h-rules
- **vlines** (array): 额外竖线，每项为 (col: int, start: int, end: none|int, stroke: stroke) / Extra v-rules
- **label** (none | label): 表格标签，用于交叉引用 / Label for cross-reference
- **..extra-args**: 任何其它命名参数（如 fill, stroke, gutter, column-gutter, row-gutter, rows）会原样透传给底层 table(...), 与 tablem 的高级用法保持一致。stroke 用户传值时会覆盖我们三线模式默认的 stroke: none（建议这种情况下同时设 three-line-table: false, 否则三条手画 hline 会叠在用户 stroke 之上）。Any other named args (e.g. fill, stroke, gutter, column-gutter, row-gutter, rows) are passed through to the underlying table(...), mirroring tablem's advanced-usage surface. A user-provided stroke overrides the stroke: none we use in three-line mode (consider also passing three-line-table: false, otherwise the three manual hlines stack on top of user stroke).
- **content** (content): Markdown 风格的表格内容 / Markdown-style table content

Parameters

```

captab(
  columns,
  cols,
  align,
  size,
  leading,
  inset,
  cell-inset,
  caption,
  caption-en,
  refer-to,
  show-caption,
  caption-position,
  caption-align,
  placement,
  width,
  three-line-table,
  top-rule,
  middle-rule,
  bottom-rule,
  extra-rule,
  breakable,
  repeat-header,
  show-continued-caption,
  ,
  continued-caption,
  hlines,
  vlines,
  label,
  ..extra-args,
  content
)

```

IX.4 Note Module (*note.typ*)

The note module provides table and figure notes with auto-width matching and multiple width modes.

- [capfnote\(\)](#)
- [captnote\(\)](#)

IX.4.1 capfnote

创建图片注释(图注), 从 `figure-style-config` 读取样式, 默认宽度自动匹配。Create a figure note; reads style from `figure-style-config`, auto-matches figure width.

- `width` (auto | ratio | length): 图注宽度 / Note width
- `justify` (auto | bool): 是否两端对齐 / Text justification

- `content (content)`: 图注内容 / Note content

Parameters

```
capfnote(
  width,
  justify,
  content
)
```

IX.4.2 captnote

创建表格注释（表注），默认宽度自动匹配 `captab` 表格宽度。Create a table note; default width auto-matches `captab` width.

- `width (auto | ratio | length)`: 表注宽度 / Note width
- `justify (auto | bool)`: 是否两端对齐（auto=使用全局配置） / Text justification
- `content (content)`: 表注内容 / Note content

Parameters

```
captnote(
  width,
  justify,
  content
)
```

IX.5 Figure Module (*figure.typ*)

The figure module provides single figure and multi-subfigure layout functions, supporting bilingual captions, overlay labels, and subcaptions.

- `capfig()`
- `capsubfig()`

IX.5.1 capfig

创建带双语标题的单个图片，标题始终在图片下方。Create a single figure with bilingual caption; caption always below image.

- `content (content)`: 图片内容（通常为 `image()` 调用） / Figure content (usually `image()`)
- `caption (none | content)`: 主语言标题 / Main language caption
- `caption-en (none | content)`: 英文标题 / English caption
- `label (none | label)`: 图片标签，用于交叉引用 / Figure label for cross-references
- `refer-to (none | label)`: 续图引用的原图标签 / Original label for continued figure
- `show-caption (auto | bool)`: 续图是否显示标题文本 / Show caption in continued figure
- `figure-above (auto | length)`: 图片上方间距 / Space above figure

- `figure-below` (auto | length): 图片下方间距 / Space below figure
- `caption-above` (auto | length): 标题与图片间距 / Space between image and caption
- `caption-leading` (auto | length): 标题行距 / Caption line spacing

Parameters

```
capfig(
    content,
    caption,
    caption-en,
    label,
    refer-to,
    show-caption,
    figure-above,
    figure-below,
    caption-above,
    caption-leading,
    placement
)
```

IX.5.2 capsubfig

创建多子图网格布局，支持子标题、覆盖标签和主双语标题。 Create a multi-subfigure grid layout with subcaptions, overlay labels, and bilingual main caption.

每个子图为字典: (`content: ...`, `subcaption: ...`, `label: ...`, `label-style-override: (...)`)。 Each subfigure is a dict: (`content: ...`, `subcaption: ...`, `label: ...`, `label-style-override: (...)`).

- `subfigs` (array): 子图字典数组 / Array of subfigure dicts
- `columns` (auto | int): 每行列数, `auto`=所有子图在一行 / Columns per row
- `caption` (none | content): 主标题 (主语言) / Main caption (main language)
- `caption-en` (none | content): 主标题 (英文) / Main caption (English)
- `label` (none | label): 主图标签 / Main figure label
- `refer-to` (none | label): 续图引用的原图标签 / Original label for continuation
- `show-caption` (auto | bool): 续图是否显示标题 / Show caption in continuation
- `gutter` (auto | length): 子图水平间距 / Horizontal gap between subfigures
- `subcaption-pos` (auto | str): 子标题位置 “top”/“bottom”
- `show-subcaption` (auto | bool): 是否显示子标题 / Show subcaptions
- `show-subcaption-label` (auto | bool): 子标题是否含标签 / Subcaption includes label
- `align` (auto | str): 垂直对齐 “top”/“horizon”/“bottom”
- `label-mode` (auto | none | str): 标签模式 / Label mode (none or “overlay”)
- `label-style` (auto | str): 标签样式, 如 “(a)”/“图 a” / Label style
- `label-font` (auto | array): 标签字体 / Label font

- `label-size` (auto | length): 标签大小 / Label size
- `label-offset` (auto | tuple): 标签偏移 (dx, dy) / Label offset
- `label-text-color` (auto | color): 标签颜色 / Label text color
- `label-stroke` (auto | none | stroke): 标签描边 / Label stroke
- `label-bg` (auto | none | color): 标签背景 / Label background
- `label-bg-shape` (auto | str): 背景形状 / Background shape
- `label-bg-radius` (auto | length): 背景圆角 / Background radius
- `label-bg-inset` (auto | length): 背景内边距 / Background padding
- `label-sep` (auto | str): 子图引用分隔符 / Subfigure reference separator
- `figure-above` (auto | length): 图片上方间距 / Space above figure
- `figure-below` (auto | length): 图片下方间距 / Space below figure
- `caption-above` (auto | length): 主标题上方间距 / Space above main caption
- `subcaption-above` (auto | length): 子标题上方间距 / Space above subcaption
- `subcaption-below` (auto | length): 子标题下方间距 / Space below subcaption

Parameters

```
capsubfig(  
    subfigs,  
    columns,  
    caption,  
    caption-en,  
    label,  
    refer-to,  
    show-caption,  
    gutter,  
    subcaption-pos,  
    show-subcaption,  
    show-subcaption-label,  
    align,  
    label-mode,  
    label-style,  
    label-font,  
    label-size,  
    label-offset,  
    label-text-color,  
    label-stroke,  
    label-bg,  
    label-bg-shape,  
    label-bg-radius,  
    label-bg-inset,  
    label-sep,  
    subref-style,  
    figure-above,  
    figure-below,  
    caption-above,  
    subcaption-above,  
    subcaption-below,  
    subcaption-number-title-spacing,  
    placement  
)
```

Part X

Frequently Asked Questions

X.1 Why Markdown syntax for tables?

The Markdown table syntax (via `tablem`) is:

- Familiar to most users
- Easy to read and edit
- Quick to type
- Compatible with many editors

X.2 How do I create tables without captions?

Simply omit the `caption` parameter:

```
1 #captab()[
2   | A | B |
3   | - | - |
4   | 1 | 2 |
5 ]
```

X.3 Can I use regular Typst table syntax?

Yes! Use `bicap` to add a caption to any native Typst table:

```
1 #align(center)[
2   #bicap(
3     caption: [Regular Typst Table],
4     caption-en: [Regular Typst Table],
5     kind: "table",
6   )
7   #table(
8     columns: 3,
9     [A], [B], [C],
10    [1], [2], [3],
11  )
12 ]
```

Table 20: Regular Typst Table

A	B	C
1	2	3

X.4 How do I reference tables and figures?

Use Typst's standard label and reference system:

```
1 #captab(caption: [...], label: <tab:example>)[...]  
2  
3 See @tab:example for details.
```

X.5 Why doesn't my continued table show the caption?

This applies specifically to manual `refer-to` mode: by default, when `caption` is omitted (and `show-caption` is `auto`), the continuation only shows the “Cont. Table X.Y” number. To force the caption text:

```
1 #captab(  
2   refer-to: <tab:original>,  
3   caption: [Original Caption], // provide caption text  
4   show-caption: true,          // or force it  
5 ) [...]
```

If you want the continuation caption to appear automatically on every continuation page when the table breaks, use the new `continued-caption` mode (since 0.1.0):

```
1 #captab(  
2   caption: [Original Caption],  
3   continued-caption: true,    // emits "Cont. Table X.Y Caption" on each continuation page  
4 ) [...]
```

See [Tables in Detail](#) → [Continued Tables](#) for the full comparison.

X.6 How do I fix missing indentation after tables in Chinese?

The package automatically detects the language and fixes indentation (for Chinese, Japanese, etc.).

For manual control:

```
1 #show: captab-style.with(  
2   after-indent: true, // Force fix  
3   // or  
4   after-indent: false, // Disable fix  
5 )
```

X.7 How do I show bilingual captions in the table of contents?

```

1 #show: captab-style.with(
2   outline-bilingual: true,      // Enable bilingual outline
3   outline-separator: " / ",    // Separator
4   outline-newline: false,      // true=newline, false=same line
5 )

```

X.8 Why no `capsubtab` (sub-tables)?

`cap-able` currently does not ship a `capsubtab`. Reasons:

1. Sub-tables are rare in academic typesetting. Most style guides do not have a dedicated section on sub-tables; the prevailing practice is to merge comparison data into a single larger table with an extra “Group” column rather than splitting into (a)(b) sub-tables. Sub-figures (`capsubfig`) are everywhere in the literature; sub-tables show up in only a small fraction of cases.
2. Page-break behaviour is harder. A side-by-side row of sub-tables is an atomic block — long content overflows rather than flowing across pages; sub-figures (typically images) are short enough that this rarely matters.
3. A workable manual recipe already exists. For two small tables side by side, `grid + multiple captab(caption: none)` wrapped in an outer `figure(kind: table, ...)` is enough; the only thing missing is automatic sub-table cross-references like `@tab:sub-a → “1.2a”`, and most authors just write “as shown in Table 1.2(a)” inline without a label anyway.

Status: an issue has been opened upstream to track this. No requests to date. If you actually need it and the manual `grid` recipe doesn’t fit, please +1 the GitHub issue and we’ll implement when demand materialises.

Example recipe for side-by-side small tables:

```

1 #figure(
2   kind: table,
3   supplement: [Table],
4   caption: figure.caption(position: top)[Experimental vs. control group data],
5   grid(
6     columns: 2,
7     column-gutter: 1em,
8     align: top,
9     [
10      *(a) Experimental* \

```



```
11      #captab(caption: none)[
12      | ID | Value |
13      | -- | ----- |
14      | 1 | 100 |
15      | 2 | 200 |
16      ]
17  ],
18  [
19      *(b) Control* \
20      #captab(caption: none)[
21      | ID | Value |
22      | -- | ----- |
23      | 1 | 95 |
24      | 2 | 205 |
25      ]
26  ],
27  ),
28  )<tab:cmp>
```

Table 21: Experimental vs. control group data	
(a) Experimental	(b) Control
ID Value	ID Value
1 100	1 95
2 200	2 205

Reference: `@tab:cmp` resolves to “Table 21”. Sub-tables (a)(b) get no automatic labels; just write “as shown in Table 21(a)” inline.

X.9 Why does a smaller number appear below a larger one after using `placement`?

When you float an earlier-declared table to the page bottom with `placement: bottom`, the page may read “Table 2 ... Table 1” top-to-bottom — the smaller number sitting below the larger.

This is not a bug — it is the inherent behavior of floats, and LaTeX behaves identically (a `\begin{table}[b]` Table 1 likewise lands below a later in-flow Table 2). Floating by definition decouples physical position from number order:

- The number always follows source declaration order — this is the academic standard, and in-text references like “see Table 1” rely on it.
- The physical position follows the `placement` target.

X.9.1 If you want “the higher table to have the smaller number”

You can control this yourself, with no `cap-able` setting changes: since numbering follows declaration order, simply declare the table you want numbered first first.

```
1 // Desired: the upper table = Table 1, the bottom-floated table = Table 2
2 #captab(caption: [the upper one])[ ... ] // declared 1st → Table 1
3 #captab(caption: [floated to bottom], placement: bottom)[ ... ] // declared 2nd → Table 2
```

Declare the physically-higher table first → it gets the smaller number; declare the one to be floated later and add `placement: bottom`. For same-page floats this gives 100% control.

⚠ Do not manually manipulate the counter. Do not try to force numbers via `counter(figure.where(kind: table)).update(...)` or similar — `cap-able` registers numbers through an internal hidden-figure mechanism (a `+1 / -1 / step` sequence); manual counter edits fight that machinery and will desync numbering or cross-references. Reordering your `captab` declarations is the only correct fix.

Part XI

Changelog

XI.1 Version 0.1.2

Fixed:

- `captab-style / capfig-style` could trigger Typst's "layout did not converge within 5 attempts" warning when re-applied inside a nested `context`. Root cause: the patch implementation used a `context { state.get(); modify; state.update(new) }` read-then-write, and Typst's layout convergence checker would treat each iteration as a possible state change and fail to converge. Replaced with the functional form `state.update(prev => modify(prev))` — the updater is a pure function and does not read state, so Typst can judge idempotence directly. The PDF output was already correct; the warning was emitted on every compile. Reported downstream in a thesis template.
- With `use-chapter: true`, the body caption's chapter prefix could intermittently render as `0` (e.g. "Table 0.1" instead of "Table 3.1") while the LoT and `@ref` still showed the correct number. Root cause: the body caption read `counter(heading).get()` at its own `context { ... }` render position, which could drift to `(0,)` across layout iterations; the LoT / `@ref` path went through the figure's own `numbering: closure`, which Typst evaluates at the figure's landed location and is therefore stable. Fixed by having the body caption also query the hidden figure's label to obtain its location and read `counter(heading).at(loc)` — matching the LoT path. An anchor label `__bicap_anchor_<n>` is auto-synthesised when the user did not supply one.
- The caption could be separated from its table/figure across a page break (issue 16). The `breakable: true` outer block allowed a page break to fall between the caption and the body, orphaning the caption at the bottom of one page while the body started on the next. The "leading" block (the caption for `caption-position: top`, the body for `bottom`) is now wrapped in `block(sticky: true)`, so it moves to the next page together with the content it leads — never separated. The body still breaks internally as normal.
- `placement: top / bottom / auto` no longer breaks the `figure-above / figure-below` spacing (and `captab's caption-above / table-below`) (issue 14). A floated element is wrapped in `place(float: true)`, on which outer `v()` / block margins have no effect; the float-to-body spacing now goes through `place's clearance` parameter. A float only has one meaningful gap (toward the body): `top` uses the below spacing, `bottom` uses the above spacing, `auto` falls back to below.

Added / Deprecated:

- `continued-caption` renamed to `show-continued-caption` (consistent with `show-subcaption / show-subcaption-label`). The old name `continued-caption` still works in 0.1.x and will be

removed in 0.2.0. The preferred name wins if both are passed. Affects `captab / captab-style / bicap`.

XI.2 Version 0.1.1

Fixed:

- User-supplied `hlines` row index was incorrectly shifted relative to the markdown table when `continued-caption: true` (issue 8). The injected caption row at `y=0` was not compensated for in the user's row, so `row: 2` ended up between the caption and the header. After the fix, row is always indexed against the markdown table itself, regardless of `continued-caption`.  Compatibility: if you used `row: N+1` as a workaround, please switch back to `row: N` — otherwise you'll get two overlapping lines.
- `subcaption-number-title-spacing` with a length value (e.g. `0.3em`) raised "cannot join string with length" because the length was being joined into the content as a value; this also suppressed the hidden-figure registration, leading to a chain failure where `@subfig` reported "label not exist" (issue 9). The fix routes the value through `handle-spacing`, which wraps lengths/relatives into `h(...)` and passes content through.

Added:

- `subref-style` config on `capfig-style / capsubfig` (default "letter", new "full") — controls the letter style of `@subfig` cross-references. "letter" shows just the letter (backward-compatible, Fig. 1a); "full" keeps the `label-style` decorations (Fig. 1(a)). In "full" mode `label-sep` defaults to empty (decorations already separate visually). Issue 10
- `extra-rule` config on `captab / captab-style` (default `0.5pt`) — default stroke for `hlines / vlines` entries that omit stroke, so you don't repeat the same stroke per line. Accepts a single value (shared by `h` & `v`) or a `(h: ..., v: ...)` dict for per-axis. Per-line stroke still wins.
- `hlines / vlines` entries now accept an int shorthand: `hlines: (2, 3, 4)` is equivalent to `((row: 2,), (row: 3,), (row: 4,))`. Can mix with full dicts: `hlines: (2, (row: 5, stroke: 1pt), 7)`.

XI.3 Version 0.1.0

New features:

- `captab` and `captab-style` gain a `breakable` parameter (bool, default `true`) — controls whether the three-line table can break across pages. The outer `block` now follows this flag, so long tables flow naturally instead of being clipped to one page.
- New `repeat-header` parameter (bool / positive int, default `true`) — repeats the markdown header row on each continuation page using Typst's native `table.header(repeat: ...)`. `true` = repeat on every continuation; `n` = first `n` continuation pages only; `false` = disable.

- New `continued-caption` parameter (bool, default `false`) — adds a “Cont. Table X.Y caption” header on each continuation page (same format as `refer-to` mode; number anchored to the main table). The caption row and markdown header row live in two separate `table.header` blocks (with `level: 1/2`), so `continued-caption: true` can coexist with `repeat-header: false`. Continuation cells locate the main table via `query(label)` and delegate to the `refer-to` branch of `_make_caption_content`, so they never re-register a figure or advance the counter; `cap-able` auto-synthesises a hidden label when the user did not provide one.
- `bicap` now supports the `#bicap(...)[body]` calling form: wraps caption + body in a single non-breakable block, conceptually a `cap-able-flavoured figure(body, caption: ...)`. `body` can also be passed via the named arg `body: [...]`.
- `bicap` gains `continued-caption` (bool, default `false`): when `kind: "table"`, every native `#table()` in `body` has its caption repeated on each continuation page (same mechanism as `captab(continued-caption: true)`). Convention: one table per `bicap`; multiple tables share the same injected caption.
- `bicap` gains `repeat-header` (auto/bool/int, default `auto`): overrides the `repeat` setting of the markdown header inside `body's #table()`. Independent of `continued-caption`; e.g. `repeat-header: false` to disable header repetition while keeping caption repetition (or alone).
- `captab` and `captab-style` gain `three-line-table` (bool, default `true`): set to `false` to drop the three-line style and use Typst's default `table()` grid stroke instead. `top-rule / middle-rule / bottom-rule` are inactive in that mode; user `hlines / vlines` still work.
- `captab` adds an explicit `align: auto` parameter and a `..extra-args` sink that forwards any other named arguments (`fill / stroke / gutter / column-gutter / row-gutter / rows`, etc.) straight to the underlying `table(...)`, matching `tablem's` advanced-usage surface. A user-supplied `stroke` overrides the `stroke: none` we emit in three-line mode (combine with `three-line-table: false` for a clean grid; otherwise the three manual `hlines` stack on top of your stroke).
- `cell-inset` and `inset` are now aliases of each other across `captab / captab-style`: previously `captab` only accepted `inset`: while the global config only accepted `cell-inset:`. Both names now work everywhere; if both are passed, `cell-inset` wins (matches the canonical state key).
- Table-width system reworked. Default changed from “fill 100% of text width” to `auto` (sized to content). New `width` parameter on `captab` and `captab-style`, accepting `auto / <length> (e.g. 8cm) / <ratio> (e.g. 80%, 80.5%)`. `set-table-width` gains a `width: parameter` (legacy `percentage: int` still supported and converted to `<int>%`). Priority: `per-call > global state > auto`. When `width != auto` and the user did not pass `columns`, the default switches to `(1fr,) * N` so the table fills the configured width; when `width == auto`, the default is `(auto,) * N` for content sizing.
- Fix `captab's` alignment handling: passing an already-2D alignment (e.g. `align: horizon + center`) no longer panics with “cannot add a vertical and a 2D alignment”. The logic now only appends `+ horizon` when the alignment's `y` component is missing.

- New `caption-position` config (`top` / `bottom`) — controls whether the caption is above or below body for both `captab` and `bicap()`[`body`]. Defaults follow `kind: table=top, figure=bottom`. Override globally via `captab-style` / `capfig-style` / `cap-style`, or per call via `captab(caption-position: ...)` / `bicap(position: ...)`. Known limitation: `caption-position: bottom` and `continued-caption: true` are incompatible (continuation repetition would require `table.footer`, which locks the caption inside the table's column boundaries); when both are set, `continued-caption` is silently disabled.
- New `caption-align` config (default `"center"`) — controls horizontal alignment of the caption. Five values: `"center"` / `"left"` / `"right"` (`table-local`) / `"text-left"` / `"text-right"` (`text-area-local`). Also accepts a dict (`main: ...`, `continued: ...`) to split main vs continuation captions (missing keys fall back to `"center"`). Exposed via `cap-style` / `captab-style` / `capfig-style` / `captab` / `bicap`. Known limitation: continuation captions live inside `table.header` and are locked to the table's column width — `"text-left"` / `"text-right"` on the continuation side silently degrade to `"left"` / `"right"`; for true text-anchored continuation captions, fall back to manual `refer-to` splitting.
- Renamed `repeat-caption` → `continued-caption` (no alias retained, since 0.1.0 has not been merged yet). Affects both `captab` and `bicap`. The semantics are unchanged — `bool`, default `false`, controls whether the caption is auto-repeated on each continuation page.
- Fixed `numbering-format` being ignored when `use-chapter: false` — that branch hard-coded `numbering("1", num)`, so user-set formats like `"(A)"` or `"App. 1"` had no effect. Both branches now honour the configured format.
- New `placement` config (default `none`) — mirrors Typst's native `figure(placement: ...)`. Four values: `none` (in-flow, default) / `top` / `bottom` (float to top/bottom of current or next page) / `auto` (Typst picks `top` or `bottom` based on which edge is closer to the current position). Implemented by wrapping the outer block in `place(float: true)`; the hidden figure (cross-ref anchor) rides along inside the float wrapper, so `@ref` resolves to the float's landing page. Exposed via `cap-style` / `captab-style` / `capfig-style` / `captab` / `capfig` / `capsubfig` / `bicap`. Known limitations: (1) floats force `breakable: false` (Typst doesn't allow floats to break across pages); (2) tables taller than a page will overflow when floated — keep `placement: none` for them; (3) incompatible with `continued-caption: true` (which depends on page breaks).
- New `caption-text` config (default `(:)`) — pass-through dict for the caption's `text(...)` rendering. Accepts any Typst native `text()` parameter (`font` / `fill` / `tracking` / `spacing` / `stretch` / ...). Two forms: a flat dict (`(font: "Times")`) applies to the whole caption; a nested dict (containing any of `whole` / `prefix` / `supplement` / `number` / `body`) layers per part with inner overriding outer. Solves the “show rule on figure.caption doesn't change the font” problem — `cap-able`'s caption isn't a real `figure.caption` element (it's manually rendered to support bilingual layout), so `font/fill/tracking` changes must go through `caption-text`. Exposed on `cap-style` / `captab-style` / `capfig-style`; applies to continuation captions as well.

- `numbering-format` now accepts `str | function`, matching the first argument of Typst's native `numbering()`. The function form receives all heading levels + the figure/table number (when `use-chapter: true`) or just the figure/table number (when `use-chapter: false`); the user function decides how to consume them. `calculate-chapter-levels` returns 0 for function-form formats (chapter-level is meaningless there). Also switched the format-string scan to grapheme clusters (`.clusters()`) so multi-byte literals like "附 1" no longer panic on UTF-8 boundaries.
- `bicap` gains a `breakable` parameter (bool, default `true`) — whether the outer block may break across pages. The previous design locked caption + body inside `breakable: false` and prevented an inherently breakable body (e.g. a long captab) from flowing across pages; the new default is transparent. Pass `breakable: false` explicitly to restore the old “atomic caption + body” behaviour.
- `Subfigure label-style-override` dict expanded with four new keys: `style / font / size / offset`, allowing per-subfig override of the overlay label's format string, font, size and offset. `style` also propagates to cross-references — when `ref` follows overlay (subcaption hidden), `@ref` renders that subfig's own format so the letter matches what's stamped on the figure. The subcaption side does not participate in per-subfig overrides (to keep subcaptions visually uniform).
- `Subfigure label-style` now accepts `str | dict` — passing a string (e.g. `"(a)"`) keeps overlay label and subcaption prefix coupled (default behaviour); passing a dict (`overlay: "(a)", subcaption: "(a)"`) decouples them, with missing keys falling back to the package default `"(a)"`. Cross-reference letters follow whichever side is actually visible — subcaption when shown (with `show-subcaption-label: true`), else overlay (when `label-mode: "overlay"`), with subcaption as the defensive fallback when neither is visible — so the rendered `@ref` letter always matches what the reader sees.
- New `subcaption-number-title-spacing` config (default `auto`) — controls the gap between the subcaption number prefix and its body. `auto` inherits the main caption's `number-title-spacing` so subcaptions render with the same separator as the main caption (instead of the previous bare single space). Configure globally via `capfig-style(subcaption-number-title-spacing: ...)` or per call via `capsubfig(subcaption-number-title-spacing: ...)`.

Compatibility:

- Default change: `breakable` defaults to `true` from 0.1.0 onward (versus the implicit `false` of 0.0.x). To preserve the old atomic behavior, pass `breakable: false` explicitly or set it globally via `captab-style.with(...)`.
- Default change: table width defaults to `auto` (content-sized) instead of “fill 100% of text width”. To restore the old behaviour, pass `set-table-width(width: 100%)` or `captab-style.with(width: 100%)`. Tables that explicitly use `fr` columns continue to expand by their `fr` factors (constrained by the parent block); tables that previously relied on the implicit 100% width with no `columns` will now appear content-sized.

XI.4 Version 0.0.2

Bug fixes:

- `captab-style` / `capfig-style` / `cap-style` switched to patch semantics. Every parameter now defaults to `auto`, and only the fields you explicitly pass get written back to state. This means you can call `captab-style.with(...)` repeatedly to override one dimension at a time (e.g. enabling `outline-bilingual`) without resetting omitted fields (e.g. `cell-inset`) to their function defaults. `Show` / `set` rules now read the latest merged values via `context { state.get() }` at render time.

New features:

- `captab` and `captab-style` gain `top-rule` / `middle-rule` / `bottom-rule` parameters for customising the three-line table's top, middle, and bottom rules. Accept any Typst stroke value, e.g. `2pt + red, stroke(thickness: 1pt, dash: "dashed")`. Defaults remain `1.5pt` / `0.5pt` / `1.5pt`.
- `captab` gains a `columns` parameter as the preferred alias of `cols`, matching Typst's native `table(columns: ...)` naming. `cols` is still accepted as a back-compat alias.

Documentation updates:

- Complete parameter tables include `top-rule` / `middle-rule` / `bottom-rule` / `columns`.
- Added patch-semantics callouts before `captab-style` and `capfig-style` sections.
- Added `== Three-Line Strokes` subsection with usage examples.
- All `cols`: examples rewritten as `columns`:.

XI.5 Version 0.0.1

Initial release with:

- `captab` three-line table support
- Bilingual caption support
- Continued tables and figures
- `capsubfig` subfigure layouts
- Table and figure notes
- 25+ language localization
- RTL language support (Arabic, Hebrew, Persian, Urdu)
- Comprehensive configuration system
- Configuration functions: `cap-style` (unified), `captab-style`, `capfig-style`, `set-table-width`

Part XII

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